United States Patent Application Publication	20250279009
Kind Code	A1
Publication Date	September 04, 2025
Inventor(s)	Van Duren; Drew Foster et al.

NETWORK ASSISTED DETECT AND AVOID SYSTEMS FOR AERIAL VEHICLES

Abstract

Methods, systems, and devices for wireless communications are described. The described techniques provide for an aircraft communicating with a detect and avoid (DAA) server. The aircraft may operate in an autonomous mode. The aircraft may transmit information associated with the aircraft to the server, and the server may respond with situational information to trigger a situational response by the aircraft while the aircraft operates in a zone associated with the server. The server may communicate with unmanned aircraft service suppliers (USSs) to support DAA techniques for the aircraft by communicating (e.g., transmitting or receiving) information associated with the aircraft.

Inventors: **Van Duren; Drew Foster (Templeton, CA), Faccin; Stefano (San Ysidro, CA), Kim; Sunghoon (San Diego, CA), Zhang; Kefeng (Beijing, CN)**

Applicant: **QUALCOMM INCORPORATED** (San Diego, CA)

Family ID: **89192800**

Appl. No.: **18/859139**

Filed (or PCT Filed): **June 16, 2022**

PCT No.: **PCT/CN2022/099113**

Publication Classification

Int. Cl.: **G08G5/80** (20250101); **G08G5/25** (20250101)

U.S. Cl.:

CPC **G08G5/80** (20250101); **G08G5/25** (20250101);

Background/Summary

CROSS REFERENCE [0001] The present Application is a 371 national stage filing of International PCT Application No. PCT/CN2022/099113 by Van Duren et al. entitled “NETWORK ASSISTED DETECT AND AVOID SYSTEMS FOR AERIAL VEHICLES,” filed Jun. 16, 2022, which is assigned to the assignee hereof, and which is expressly incorporated by reference in its entirety herein.

FIELD OF TECHNOLOGY

[0002] The following relates to wireless communications, including network assisted detect and avoid systems for aerial vehicles.

BACKGROUND

[0003] Wireless communications systems are widely deployed to provide various types of communication content such as voice, video, packet data, messaging, broadcast, and so on. These systems may be capable of supporting communication with multiple users by sharing the available system resources (e.g., time, frequency, and power). Examples of such multiple-access systems include fourth generation (4G) systems such as Long Term Evolution (LTE) systems, LTE-Advanced (LTE-A) systems, or LTE-A Pro systems, and fifth generation (5G) systems which may be referred to as New Radio (NR) systems. These systems may employ technologies such as code division multiple access (CDMA), time division multiple access (TDMA), frequency division multiple access (FDMA), orthogonal FDMA (OFDMA), or discrete Fourier transform spread orthogonal frequency division multiplexing (DFT-S-OFDM). A wireless multiple-access communications system may include one or more base stations, each supporting wireless communication for communication devices, which may be known as user equipment (UE).

[0004] Unmanned aerial vehicles (UAVs) may implement detect and avoid (DAA) systems and techniques that support collision or conflict avoidance. In some jurisdictions, these DAA systems are required for flight operations that do not have human observers or may lack ground-based observation systems along a flight path. DAA regulations for UAVs are being written, but current solutions rely on a remote human pilot and/or solely on on-board sensors and aerial awareness of surrounding traffic.

SUMMARY

[0005] The described techniques relate to improved methods, systems, devices, and apparatuses that support network assisted detect and avoid systems for aerial vehicles. For example, the described techniques provide for an aircraft communicating with a detect and avoid (DAA) server. The aircraft may transmit information associated with the aircraft, and the server may respond with situational information to trigger a situational response by the aircraft. The server may communicate with unmanned aircraft service suppliers (USSs) to support DAA techniques for the aircraft.

[0006] A method for wireless communications at a server is described. The method may include receiving, via a radio access network and from an aircraft, first information associated with the aircraft, transmitting, to the aircraft in response to receiving the first information, situational information to trigger a situational response by the aircraft while the aircraft operates in a zone associated with the server, and communicating, between the server and an unmanned aircraft service supplier, second information associated with the aircraft, the communicating being via a network function and based on receiving the first information.

[0007] An apparatus for wireless communications at a server is described. The apparatus may include a processor, memory coupled with the processor, and instructions stored in the memory. The instructions may be executable by the processor to cause the apparatus to receive, via a radio access network and from an aircraft, first information associated with the aircraft, transmit, to the

aircraft in response to receiving the first information, situational information to trigger a situational response by the aircraft while the aircraft operates in a zone associated with the server, and communicate, between the server and an unmanned aircraft service supplier, second information associated with the aircraft, the communicating being via a network function and based on receiving the first information.
[0008] Another apparatus for wireless communications at a server is described. The apparatus may include means for receiving, via a radio access network and from an aircraft, first information associated with the aircraft, means for transmitting, to the aircraft in response to receiving the first information, situational information to trigger a situational response by the aircraft while the aircraft operates in a zone associated with the server, and means for communicating, between the server and an unmanned aircraft service supplier, second information associated with the aircraft, the communicating being via a network function and based on receiving the first information.
[0009] A non-transitory computer-readable medium storing code for wireless communications at a server is described. The code may include instructions executable by a processor to receive, via a radio access network and from an aircraft, first information associated with the aircraft, transmit, to the aircraft in response to receiving the first information, situational information to trigger a situational response by the aircraft while the aircraft operates in a zone associated with the server, and communicate, between the server and an unmanned aircraft service supplier, second information associated with the aircraft, the communicating being via a network function and based on receiving the first information.
[0010] In some examples of the method, apparatuses, and non-transitory computer-readable medium described herein, transmitting the situational information may include operations, features, means, or instructions for transmitting a warning indicative of a potential conflict associated with the aircraft.
[0011] In some examples of the method, apparatuses, and non-transitory computer-readable medium described herein, transmitting the situational information may include operations, features, means, or instructions for transmitting an indication of an avoidance instruction for avoiding a potential conflict by the aircraft.
[0012] In some examples of the method, apparatuses, and non-transitory computer-readable medium described herein, transmitting the situational information may include operations, features, means, or instructions for transmitting neighboring vehicle information associated with one or more vehicles operating in the zone or another proximal zone.
[0013] In some examples of the method, apparatuses, and non-transitory computer-readable medium described herein, transmitting the situational information may include operations, features, means, or instructions for transmit topographical information, ground-based obstruction information, or both associated with the zone or another proximal zone.
[0014] In some examples of the method, apparatuses, and non-transitory computer-readable medium described herein, receiving the first information may include operations, features, means, or instructions for receiving a deconfliction request or conflict warning associated with a potential conflict detected by the aircraft, where situational information may be transmitted based on receiving the deconfliction request or the conflict warning and includes an avoidance instruction.
[0015] In some examples of the method, apparatuses, and non-transitory computer-readable medium described herein, receiving the first information may include operations, features, means, or instructions for receiving metadata associated with the aircraft, where the metadata includes an aircraft type, an aircraft capability, an aircraft position, an aircraft velocity, an aircraft identity, aircraft operator information, or a combination thereof.
[0016] Some examples of the method, apparatuses, and non-transitory computer-readable medium described herein may further include operations, features, means, or instructions for receiving, from the aircraft, third information associated with one or more vehicles detected by the aircraft.
[0017] In some examples of the method, apparatuses, and non-transitory computer-readable

medium described herein, the one or more vehicles include a piloted aircraft, an aircraft operating in the autonomous flying mode, a ground vehicle, or a combination thereof.
[0018] In some examples of the method, apparatuses, and non-transitory computer-readable medium described herein, communicating the second information may include operations, features, means, or instructions for transmitting, to the unmanned aircraft service supplier, an indication of conflict information associated with the aircraft, an indication of vehicle congestion information of the zone, or a combination thereof.
[0019] In some examples of the method, apparatuses, and non-transitory computer-readable medium described herein, communicating the second information may include operations, features, means, or instructions for transmitting, to the unmanned aircraft service supplier, a request for the second information associated with the aircraft and receiving, from the unmanned aircraft service supplier, in response to transmitting the request, the second information.
[0020] Some examples of the method, apparatuses, and non-transitory computer-readable medium described herein may further include operations, features, means, or instructions for determining, based on receiving the first information, that a potential conflict exists for the aircraft, a second aircraft, or both and transmitting, to the aircraft, the second aircraft, or both, an indication of the potential conflict.
[0021] Some examples of the method, apparatuses, and non-transitory computer-readable medium described herein may further include operations, features, means, or instructions for receiving, from one or more sensors associated with the server, zone information associated with the zone, where the situational information may be based on the zone information.
[0022] In some examples of the method, apparatuses, and non-transitory computer-readable medium described herein, the one or more sensors include a detect and avoid broadcast receiver, a broadcast remote identifier receiver, an automatic dependent surveillance-broadcast receiver, a weather sensor, a radar, a new-radio sensor, a lidar, an aircraft transponder, or a combination thereof.
[0023] In some examples of the method, apparatuses, and non-transitory computer-readable medium described herein, the server communicates with the aircraft via an access link, a sidelink, or both.
[0024] A method for wireless communications a first aircraft is described. The method may include receiving first information associated with one or more second vehicles, transmitting, to a server operating in a radio access network and based on receiving the first information associated with the one or more second vehicles, second information associated with the first aircraft, and receiving, from the server based on transmitting the second information, situational information to trigger a situational response by the first aircraft while the first aircraft operates in a zone associated with the server.
[0025] An apparatus for wireless communications a first aircraft is described. The apparatus may include a processor, memory coupled with the processor, and instructions stored in the memory. The instructions may be executable by the processor to cause the apparatus to receive first information associated with one or more second vehicles, transmit, to a server operating in a radio access network and based on receiving the first information associated with the one or more second vehicles, second information associated with the first aircraft, and receive, from the server based on transmitting the second information, situational information to trigger a situational response by the first aircraft while the first aircraft operates in a zone associated with the server.
[0026] Another apparatus for wireless communications a first aircraft is described. The apparatus may include means for receiving first information associated with one or more second vehicles, means for transmitting, to a server operating in a radio access network and based on receiving the first information associated with the one or more second vehicles, second information associated with the first aircraft, and means for receiving, from the server based on transmitting the second information, situational information to trigger a situational response by the first aircraft while the

first aircraft operates in a zone associated with the server.

[0027] A non-transitory computer-readable medium storing code for wireless communications a first aircraft is described. The code may include instructions executable by a processor to receive first information associated with one or more second vehicles, transmit, to a server operating in a radio access network and based on receiving the first information associated with the one or more second vehicles, second information associated with the first aircraft, and receive, from the server based on transmitting the second information, situational information to trigger a situational response by the first aircraft while the first aircraft operates in a zone associated with the server.

[0028] In some examples of the method, apparatuses, and non-transitory computer-readable medium described herein, receiving the situational information may include operations, features, means, or instructions for receiving a warning indicative of a potential conflict associated with the first aircraft.

[0029] In some examples of the method, apparatuses, and non-transitory computer-readable medium described herein, receiving the situational information may include operations, features, means, or instructions for receiving an indication of an avoidance instruction for avoiding a potential conflict by the first aircraft.

[0030] In some examples of the method, apparatuses, and non-transitory computer-readable medium described herein, receiving the situational information may include operations, features, means, or instructions for receiving information associated with one or more vehicles operating in the zone or another proximal zone.

[0031] In some examples of the method, apparatuses, and non-transitory computer-readable medium described herein, receiving the situational information may include operations, features, means, or instructions for receive topographical information, ground-based obstruction information, or both associated with the zone or another proximal zone.

[0032] In some examples of the method, apparatuses, and non-transitory computer-readable medium described herein, transmitting the second information may include operations, features, means, or instructions for transmitting deconfliction request or conflict warning associated with a potential conflict detected by the first aircraft, where situational information may be received based on transmitting the deconfliction request or the conflict warning and includes an avoidance instruction.

[0033] In some examples of the method, apparatuses, and non-transitory computer-readable medium described herein, transmitting the second information may include operations, features, means, or instructions for transmitting metadata associated with the first aircraft, where the metadata includes an aircraft type, an aircraft capability, an aircraft position, an aircraft velocity, an aircraft identity, aircraft operator information, or a combination thereof.

[0034] Some examples of the method, apparatuses, and non-transitory computer-readable medium described herein may further include operations, features, means, or instructions for transmitting, to the server, an indication of the first information associated with one or more vehicles received by the first aircraft.

[0035] In some examples of the method, apparatuses, and non-transitory computer-readable medium described herein, the one or more second vehicles include a piloted aircraft, an aircraft operating in an autonomous flying mode, a ground vehicle, or a combination thereof.

[0036] In some examples of the method, apparatuses, and non-transitory computer-readable medium described herein, the first aircraft communicates with the server via an access link, a sidelink, or both.

[0037] Some examples of the method, apparatuses, and non-transitory computer-readable medium described herein may further include operations, features, means, or instructions for receiving, from an unmanned aircraft service supplier, an indication of reachability information for the server, an indication of the zone associated with the server, or both, where the first aircraft communicates with the server based on receiving the reachability information, the indication of the zone, or both.

[0038] A method for communications at a first unmanned aircraft service supplier is described. The method may include receiving, from a discovery and synchronization service, information associated with a second unmanned aircraft service supplier, receiving, from the second unmanned aircraft service supplier based on receiving the information, detect and avoid service area entity information associated with one or more detect and avoid servers managed by the second unmanned aircraft service supplier, and transmitting, to an aircraft, an indication of the detect and avoid service area entity information that the aircraft is use.

[0039] An apparatus for communications at a first unmanned aircraft service supplier is described. The apparatus may include a processor, memory coupled with the processor, and instructions stored in the memory. The instructions may be executable by the processor to cause the apparatus to receive, from a discovery and synchronization service, information associated with a second unmanned aircraft service supplier, receive, from the second unmanned aircraft service supplier based on receiving the information, detect and avoid service area entity information associated with one or more detect and avoid servers managed by the second unmanned aircraft service supplier, and transmit, to an aircraft, an indication of the detect and avoid service area entity information that the aircraft is use.

[0040] Another apparatus for communications at a first unmanned aircraft service supplier is described. The apparatus may include means for receiving, from a discovery and synchronization service, information associated with a second unmanned aircraft service supplier, means for receiving, from the second unmanned aircraft service supplier based on receiving the information, detect and avoid service area entity information associated with one or more detect and avoid servers managed by the second unmanned aircraft service supplier, and means for transmitting, to an aircraft, an indication of the detect and avoid service area entity information that the aircraft is use.

[0041] A non-transitory computer-readable medium storing code for communications at a first unmanned aircraft service supplier is described. The code may include instructions executable by a processor to receive, from a discovery and synchronization service, information associated with a second unmanned aircraft service supplier, receive, from the second unmanned aircraft service supplier based on receiving the information, detect and avoid service area entity information associated with one or more detect and avoid servers managed by the second unmanned aircraft service supplier, and transmit, to an aircraft, an indication of the detect and avoid service area entity information that the aircraft is use.

[0042] In some examples of the method, apparatuses, and non-transitory computer-readable medium described herein, receiving the information associated with the second unmanned aircraft service supplier may include operations, features, means, or instructions for receiving an indication of reachability information for the one or more detected and avoid servers, a location of the one or more detect avoid servers, a zone identifier associated with the one or more detect and avoid servers, or a combination thereof.

[0043] Some examples of the method, apparatuses, and non-transitory computer-readable medium described herein may further include operations, features, means, or instructions for transmitting, to the second unmanned aircraft service supplier in response to receiving the information, a request for the detect and avoid service area entity information, where the detect and avoid service area entity information may be received based on transmitting the request

---

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0044] FIG. **1** illustrates an example of a wireless communications system that supports network assisted detect and avoid systems for aerial vehicles in accordance with one or more aspects of the present disclosure.

[0045] FIG. **2** illustrates an example of a wireless communications system that supports network assisted detect and avoid systems for aerial vehicles in accordance with one or more aspects of the present disclosure.

[0046] FIG. **3** illustrates an example of an operating environment that supports network assisted detect and avoid systems for aerial vehicles in accordance with one or more aspects of the present disclosure.

[0047] FIG. **4** illustrates an example of a process flow that supports network assisted detect and avoid systems for aerial vehicles in accordance with one or more aspects of the present disclosure.

[0048] FIGS. **5** and **6** show block diagrams of devices that support network assisted detect and avoid systems for aerial vehicles in accordance with one or more aspects of the present disclosure.

[0049] FIG. **7** shows a block diagram of a communications manager that supports network assisted detect and avoid systems for aerial vehicles in accordance with one or more aspects of the present disclosure.

[0050] FIG. **8** shows a diagram of a system including a device that supports network assisted detect and avoid systems for aerial vehicles in accordance with one or more aspects of the present disclosure.

[0051] FIGS. **9** and **10** show block diagrams of devices that support network assisted detect and avoid systems for aerial vehicles in accordance with one or more aspects of the present disclosure.

[0052] FIG. **11** shows a block diagram of a communications manager that supports network assisted detect and avoid systems for aerial vehicles in accordance with one or more aspects of the present disclosure.

[0053] FIG. **12** shows a diagram of a system including a device that supports network assisted detect and avoid systems for aerial vehicles in accordance with one or more aspects of the present disclosure.

[0054] FIGS. **13** through **15** show flowcharts illustrating methods that support network assisted detect and avoid systems for aerial vehicles in accordance with one or more aspects of the present disclosure.

DETAILED DESCRIPTION

[0055] Aircrafts, such as unmanned aerial vehicles (UAVs), drones, auto-pilot assisted aircrafts, and the like, may operate in autonomous modes. The aircrafts may implement detect and avoid (DAA) systems and techniques that support traffic separation, collision or conflict avoidance. In some jurisdictions, these DAA systems are required for flight operations that do not have human observers or ground-based observation systems along a flight path. DAA regulations for UAVs are being written, but current solutions rely on a remote human pilot and/or solely on on-board sensors and aerial awareness of surrounding traffic.

[0056] Techniques described herein support ground-based network infrastructure to support DAA systems in unmanned or autonomously operating aircraft operations. To support DAA systems, a server/node (which may be referred to as a local detect and avoid server (LDS)) tailored for DAA is positioned in a radio access network. The server is configured to wirelessly communicate with aircraft and to communicate information with an unmanned aircraft service supplier (USS). The server may receive information from an aircraft, and the information may include aircraft information (e.g., identifiers, capabilities, other detected aircraft, position/velocity vectors). The server may also transmit situational information to the aircraft, and the situational information may include general information such as other aircraft identifiers or collision avoidance requests/directives, and the situational information may trigger a situational response by the aircraft. The server may communicate congestion or conflict information to the USS and receive information about the aircraft from the USS. The information received from the unmanned aircraft service supplier may include aircraft category (the category of aircraft that is in communication with the LDS server), aircraft attributes, and/or mission type (the type of flight plan or indicated intent of the aircraft that is in communication with the LDS server). The server may also communicate with various ground based sensors to further support DAA systems. These and other techniques are described in further detail with respect to the figures.

[0057] Aspects of the disclosure are initially described in the context of wireless communications systems. Aspects of the disclosure are further described with respect to a wireless communications system illustrating example communications between a aircraft, a server, and a USS to support DAA techniques, an operating environment illustrating communications for DAA service discovery and configuration, and a process flow. Aspects of the disclosure are further illustrated by and described with reference to apparatus diagrams, system diagrams, and flowcharts that relate to network assisted detect and avoid systems for aerial vehicles.

[0058] FIG. **1** illustrates an example of a wireless communications system **100** that supports network assisted detect and avoid systems for aerial vehicles in accordance with one or more aspects of the present disclosure. The wireless communications system **100** may include one or more network entities **105**, one or more UEs **115**, and a core network **130**. In some examples, the wireless communications system **100** may be a Long Term Evolution (LTE) network, an LTE-Advanced (LTE-A) network, an LTE-A Pro network, a New Radio (NR) network, or a network operating in accordance with other systems and radio technologies, including future systems and radio technologies not explicitly mentioned herein.

[0059] The network entities **105** may be dispersed throughout a geographic area to form the wireless communications system **100** and may include devices in different forms or having different capabilities. In various examples, a network entity **105** may be referred to as a network element, a mobility element, a radio access network (RAN) node, or network equipment, among other nomenclature. In some examples, network entities **105** and UEs **115** may wirelessly communicate via one or more communication links **125** (e.g., a radio frequency (RF) access link). For example, a network entity **105** may support a coverage area **110** (e.g., a geographic coverage area) over which the UEs **115** and the network entity **105** may establish one or more communication links **125**. The coverage area **110** may be an example of a geographic area over which a network entity **105** and a UE **115** may support the communication of signals according to one or more radio access technologies (RATs).

[0060] The UEs **115** may be dispersed throughout a coverage area **110** of the wireless communications system **100**, and each UE **115** may be stationary, or mobile, or both at different times. The UEs **115** may be devices in different forms or having different capabilities. Some example UEs **115** are illustrated in FIG. **1**. The UEs **115** described herein may be capable of supporting communications with various types of devices, such as other UEs **115** or network entities **105**, as shown in FIG. **1**.

[0061] As described herein, a node of the wireless communications system **100**, which may be referred to as a network node, or a wireless node, may be a network entity **105** (e.g., any network entity described herein), a UE **115** (e.g., any UE described herein), a network controller, an apparatus, a device, a computing system, one or more components, or another suitable processing entity configured to perform any of the techniques described herein. For example, a node may be a UE **115**. As another example, a node may be a network entity **105**. As another example, a first node may be configured to communicate with a second node or a third node. In one aspect of this example, the first node may be a UE **115**, the second node may be a network entity **105**, and the third node may be a UE **115**. In another aspect of this example, the first node may be a UE **115**, the second node may be a network entity **105**, and the third node may be a network entity **105**. In yet other aspects of this example, the first, second, and third nodes may be different relative to these examples. Similarly, reference to a UE **115**, network entity **105**, apparatus, device, computing system, or the like may include disclosure of the UE **115**, network entity **105**, apparatus, device, computing system, or the like being a node. For example, disclosure that a UE **115** is configured to receive information from a network entity **105** also discloses that a first node is configured to receive information from a second node.

[0062] In some examples, network entities **105** may communicate with the core network **130**, or with one another, or both. For example, network entities **105** may communicate with the core network **130** via one or more backhaul communication links **120** (e.g., in accordance with an S1, N2, N3, or other interface protocol). In some examples, network entities **105** may communicate with one another via a backhaul communication link **120** (e.g., in accordance with an X2, Xn, or other interface protocol) either directly (e.g., directly between network entities **105**) or indirectly (e.g., via a core network **130**). In some examples, network entities **105** may communicate with one another via a midhaul communication link **162** (e.g., in accordance with a midhaul interface protocol) or a fronthaul communication link **168** (e.g., in accordance with a fronthaul interface protocol), or any combination thereof. The backhaul communication links **120**, midhaul communication links **162**, or fronthaul communication links **168** may be or include one or more wired links (e.g., an electrical link, an optical fiber link), one or more wireless links (e.g., a radio link, a wireless optical link), among other examples or various combinations thereof. A UE **115** may communicate with the core network **130** via a communication link **155**.

[0063] One or more of the network entities **105** described herein may include or may be referred to as a base station **140** (e.g., a base transceiver station, a radio base station, an NR base station, an access point, a radio transceiver, a NodeB, an eNodeB (eNB), a next-generation NodeB or a giga-NodeB (either of which may be referred to as a gNB), a 5G NB, a next-generation eNB (ng-eNB), a Home NodeB, a Home eNodeB, or other suitable terminology). In some examples, a network entity **105** (e.g., a base station **140**) may be implemented in an aggregated (e.g., monolithic, standalone) base station architecture, which may be configured to utilize a protocol stack that is physically or logically integrated within a single network entity **105** (e.g., a single RAN node, such as a base station **140**).

[0064] In some examples, a network entity **105** may be implemented in a disaggregated architecture (e.g., a disaggregated base station architecture, a disaggregated RAN architecture), which may be configured to utilize a protocol stack that is physically or logically distributed among two or more network entities **105**, such as an integrated access backhaul (IAB) network, an open RAN (O-RAN) (e.g., a network configuration sponsored by the O-RAN Alliance), or a virtualized RAN (vRAN) (e.g., a cloud RAN (C-RAN)). For example, a network entity **105** may include one or more of a central unit (CU) **160**, a distributed unit (DU) **165**, a radio unit (RU) **170**, a RAN Intelligent Controller (RIC) **175** (e.g., a Near-Real Time RIC (Near-RT RIC), a Non-Real Time RIC (Non-RT RIC)), a Service Management and Orchestration (SMO) **180** system, or any combination thereof. An RU **170** may also be referred to as a radio head, a smart radio head, a remote radio head (RRH), a remote radio unit (RRU), or a transmission reception point (TRP). One or more components of the network entities **105** in a disaggregated RAN architecture may be co-located, or one or more components of the network entities **105** may be located in distributed locations (e.g., separate physical locations). In some examples, one or more network entities **105** of a disaggregated RAN architecture may be implemented as virtual units (e.g., a virtual CU (VCU), a virtual DU (VDU), a virtual RU (VRU)).

[0065] The split of functionality between a CU **160**, a DU **165**, and an RU **170** is flexible and may support different functionalities depending upon which functions (e.g., network layer functions, protocol layer functions, baseband functions, RF functions, and any combinations thereof) are performed at a CU **160**, a DU **165**, or an RU **170**. For example, a functional split of a protocol stack may be employed between a CU **160** and a DU **165** such that the CU **160** may support one or more layers of the protocol stack and the DU **165** may support one or more different layers of the protocol stack. In some examples, the CU **160** may host upper protocol layer (e.g., layer 3 (L3), layer 2 (L2)) functionality and signaling (e.g., Radio Resource Control (RRC), service data adaption protocol (SDAP), Packet Data Convergence Protocol (PDCP)). The CU **160** may be connected to one or more DUs **165** or RUs **170**, and the one or more DUs **165** or RUs **170** may host lower protocol layers, such as layer 1 (L1) (e.g., physical (PHY) layer) or L2 (e.g., radio link control (RLC) layer, medium access control (MAC) layer) functionality and signaling, and may each be at least partially controlled by the CU **160**. Additionally, or alternatively, a functional split of the protocol stack may be employed between a DU **165** and an RU **170** such that the DU **165** may support one or more layers of the protocol stack and the RU **170** may support one or more different layers of the protocol stack. The DU **165** may support one or multiple different cells (e.g., via one or more RUs **170**). In some cases, a functional split between a CU **160** and a DU **165**, or between a DU **165** and an RU **170** may be within a protocol layer (e.g., some functions for a protocol layer may be performed by one of a CU **160**, a DU **165**, or an RU **170**, while other functions of the protocol layer are performed by a different one of the CU **160**, the DU **165**, or the RU **170**). A CU **160** may be functionally split further into CU control plane (CU-CP) and CU user plane (CU-UP) functions. A CU **160** may be connected to one or more DUs **165** via a midhaul communication link **162** (e.g., F1, F1-c, F1-u), and a DU **165** may be connected to one or more RUs **170** via a fronthaul communication link **168** (e.g., open fronthaul (FH) interface). In some examples, a midhaul communication link **162** or a fronthaul communication link **168** may be implemented in accordance with an interface (e.g., a channel) between layers of a protocol stack supported by respective network entities **105** that are in communication via such communication links.

[0066] In wireless communications systems (e.g., wireless communications system **100**), infrastructure and spectral resources for radio access may support wireless backhaul link capabilities to supplement wired backhaul connections, providing an IAB network architecture (e.g., to a core network **130**). In some cases, in an IAB network, one or more network entities **105** (e.g., IAB nodes **104**) may be partially controlled by each other. One or more IAB nodes **104** may be referred to as a donor entity or an IAB donor. One or more DUs **165** or one or more RUs **170** may be partially controlled by one or more CUs **160** associated with a donor network entity **105** (e.g., a donor base station **140**). The one or more donor network entities **105** (e.g., IAB donors) may be in communication with one or more additional network entities **105** (e.g., IAB nodes **104**) via supported access and backhaul links (e.g., backhaul communication links **120**). IAB nodes **104** may include an IAB mobile termination (IAB-MT) controlled (e.g., scheduled) by DUs **165** of a coupled IAB donor. An IAB-MT may include an independent set of antennas for relay of communications with UEs **115**, or may share the same antennas (e.g., of an RU **170**) of an IAB node **104** used for access via the DU **165** of the IAB node **104** (e.g., referred to as virtual IAB-MT (vIAB-MT)). In some examples, the IAB nodes **104** may include DUs **165** that support communication links with additional entities (e.g., IAB nodes **104**, UEs **115**) within the relay chain or configuration of the access network (e.g., downstream). In such cases, one or more components of the disaggregated RAN architecture (e.g., one or more IAB nodes **104** or components of IAB nodes **104**) may be configured to operate according to the techniques described herein.

[0067] For instance, an access network (AN) or RAN may include communications between access nodes (e.g., an IAB donor), IAB nodes **104**, and one or more UEs **115**. The IAB donor may facilitate connection between the core network **130** and the AN (e.g., via a wired or wireless connection to the core network **130**). That is, an IAB donor may refer to a RAN node with a wired or wireless connection to core network **130**. The IAB donor may include a CU **160** and at least one DU **165** (e.g., and RU **170**), in which case the CU **160** may communicate with the core network **130** via an interface (e.g., a backhaul link). IAB donor and IAB nodes **104** may communicate via an F1 interface according to a protocol that defines signaling messages (e.g., an F1 AP protocol). Additionally, or alternatively, the CU **160** may communicate with the core network via an interface, which may be an example of a portion of backhaul link, and may communicate with other CUs **160** (e.g., a CU **160** associated with an alternative IAB donor) via an Xn-C interface, which may be an example of a portion of a backhaul link.

[0068] An IAB node **104** may refer to a RAN node that provides IAB functionality (e.g., access for

UEs **115**, wireless self-backhauling capabilities). A DU **165** may act as a distributed scheduling node towards child nodes associated with the IAB node **104**, and the IAB-MT may act as a scheduled node towards parent nodes associated with the IAB node **104**. That is, an IAB donor may be referred to as a parent node in communication with one or more child nodes (e.g., an IAB donor may relay transmissions for UEs through one or more other IAB nodes **104**). Additionally, or alternatively, an IAB node **104** may also be referred to as a parent node or a child node to other IAB nodes **104**, depending on the relay chain or configuration of the AN. Therefore, the IAB-MT entity of IAB nodes **104** may provide a Uu interface for a child IAB node **104** to receive signaling from a parent IAB node **104**, and the DU interface (e.g., DUs **165**) may provide a Uu interface for a parent IAB node **104** to signal to a child IAB node **104** or UE **115**.

[0069] For example, IAB node **104** may be referred to as a parent node that supports communications for a child IAB node, or referred to as a child IAB node associated with an IAB donor, or both. The IAB donor may include a CU **160** with a wired or wireless connection (e.g., a backhaul communication link **120**) to the core network **130** and may act as parent node to IAB nodes **104**. For example, the DU **165** of IAB donor may relay transmissions to UEs **115** through IAB nodes **104**, or may directly signal transmissions to a UE **115**, or both. The CU **160** of IAB donor may signal communication link establishment via an F1 interface to IAB nodes **104**, and the IAB nodes **104** may schedule transmissions (e.g., transmissions to the UEs **115** relayed from the IAB donor) through the DUs **165**. That is, data may be relayed to and from IAB nodes **104** via signaling via an NR Uu interface to MT of the IAB node **104**. Communications with IAB node **104** may be scheduled by a DU **165** of IAB donor and communications with IAB node **104** may be scheduled by DU **165** of IAB node **104**.

[0070] In the case of the techniques described herein applied in the context of a disaggregated RAN architecture, one or more components of the disaggregated RAN architecture may be configured to support network assisted detect and avoid systems for aerial vehicles as described herein. For example, some operations described as being performed by a UE **115** or a network entity **105** (e.g., a base station **140**) may additionally, or alternatively, be performed by one or more components of the disaggregated RAN architecture (e.g., IAB nodes **104**, DUs **165**, CUs **160**, RUs **170**, RIC **175**, SMO **180**).

[0071] A UE **115** may include or may be referred to as a mobile device, a wireless device, a remote device, a handheld device, or a subscriber device, or some other suitable terminology, where the “device” may also be referred to as a unit, a station, a terminal, or a client, among other examples. A UE **115** may also include or may be referred to as a personal electronic device such as a cellular phone, a personal digital assistant (PDA), a tablet computer, a laptop computer, or a personal computer. In some examples, a UE **115** may include or be referred to as a wireless local loop (WLL) station, an Internet of Things (IoT) device, an Internet of Everything (IoE) device, or a machine type communications (MTC) device, among other examples, which may be implemented in various objects such as appliances, or vehicles, meters, among other examples.

[0072] The UEs **115** described herein may be able to communicate with various types of devices, such as other UEs **115** that may sometimes act as relays as well as the network entities **105** and the network equipment including macro eNBs or gNBs, small cell eNBs or gNBs, or relay base stations, among other examples, as shown in FIG. **1**.

[0073] The UEs **115** and the network entities **105** may wirelessly communicate with one another via one or more communication links **125** (e.g., an access link) using resources associated with one or more carriers. The term “carrier” may refer to a set of RF spectrum resources having a defined physical layer structure for supporting the communication links **125**. For example, a carrier used for a communication link **125** may include a portion of a RF spectrum band (e.g., a bandwidth part (BWP)) that is operated according to one or more physical layer channels for a given radio access technology (e.g., LTE, LTE-A, LTE-A Pro, NR). Each physical layer channel may carry acquisition signaling (e.g., synchronization signals, system information), control signaling that coordinates operation for the carrier, user data, or other signaling. The wireless communications system **100** may support communication with a UE **115** using carrier aggregation or multi-carrier operation. A UE **115** may be configured with multiple downlink component carriers and one or more uplink component carriers according to a carrier aggregation configuration. Carrier aggregation may be used with both frequency division duplexing (FDD) and time division duplexing (TDD) component carriers. Communication between a network entity **105** and other devices may refer to communication between the devices and any portion (e.g., entity, sub-entity) of a network entity **105**. For example, the terms “transmitting,” “receiving,” or “communicating,” when referring to a network entity **105**, may refer to any portion of a network entity **105** (e.g., a base station **140**, a CU **160**, a DU **165**, a RU **170**) of a RAN communicating with another device (e.g., directly or via one or more other network entities **105**).

[0074] Signal waveforms transmitted via a carrier may be made up of multiple subcarriers (e.g., using multi-carrier modulation (MCM) techniques such as orthogonal frequency division multiplexing (OFDM) or discrete Fourier transform spread OFDM (DFT-S-OFDM)). In a system employing MCM techniques, a resource element may refer to resources of one symbol period (e.g., a duration of one modulation symbol) and one subcarrier, in which case the symbol period and subcarrier spacing may be inversely related. The quantity of bits carried by each resource element may depend on the modulation scheme (e.g., the order of the modulation scheme, the coding rate of the modulation scheme, or both), such that a relatively higher quantity of resource elements (e.g., in a transmission duration) and a relatively higher order of a modulation scheme may correspond to a relatively higher rate of communication. A wireless communications resource may refer to a combination of an RF spectrum resource, a time resource, and a spatial resource (e.g., a spatial layer, a beam), and the use of multiple spatial resources may increase the data rate or data integrity for communications with a UE **115**.

[0075] The time intervals for the network entities **105** or the UEs **115** may be expressed in multiples of a basic time unit which may, for example, refer to a sampling period of T.sub.s=1/(Δf.sub.max.Math.N.sub.f) seconds, for which Δf.sub.max may represent a supported subcarrier spacing, and N.sub.f may represent a supported discrete Fourier transform (DFT) size. Time intervals of a communications resource may be organized according to radio frames each having a specified duration (e.g., 10 milliseconds (ms)). Each radio frame may be identified by a system frame number (SFN) (e.g., ranging from 0 to 1023).

[0076] Each frame may include multiple consecutively-numbered subframes or slots, and each subframe or slot may have the same duration. In some examples, a frame may be divided (e.g., in the time domain) into subframes, and each subframe may be further divided into a quantity of slots. Alternatively, each frame may include a variable quantity of slots, and the quantity of slots may depend on subcarrier spacing. Each slot may include a quantity of symbol periods (e.g., depending on the length of the cyclic prefix prepended to each symbol period). In some wireless communications systems **100**, a slot may further be divided into multiple mini-slots associated with one or more symbols. Excluding the cyclic prefix, each symbol period may be associated with one or more (e.g., N.sub.f) sampling periods. The duration of a symbol period may depend on the subcarrier spacing or frequency band of operation.

[0077] A subframe, a slot, a mini-slot, or a symbol may be the smallest scheduling unit (e.g., in the time domain) of the wireless communications system **100** and may be referred to as a transmission time interval (TTI). In some examples, the TTI duration (e.g., a quantity of symbol periods in a TTI) may be variable. Additionally, or alternatively, the smallest scheduling unit of the wireless communications system **100** may be dynamically selected (e.g., in bursts of shortened TTIs (STTIs)).

[0078] Physical channels may be multiplexed for communication using a carrier according to various techniques. A physical control channel and a physical data channel may be multiplexed for signaling via a downlink carrier, for example, using one or more of time division multiplexing

(TDM) techniques, frequency division multiplexing (FDM) techniques, or hybrid TDM-FDM techniques. A control region (e.g., a control resource set (CORESET)) for a physical control channel may be defined by a set of symbol periods and may extend across the system bandwidth or a subset of the system bandwidth of the carrier. One or more control regions (e.g., CORESETs) may be configured for a set of the UEs **115**. For example, one or more of the UEs **115** may monitor or search control regions for control information according to one or more search space sets, and each search space set may include one or multiple control channel candidates in one or more aggregation levels arranged in a cascaded manner. An aggregation level for a control channel candidate may refer to an amount of control channel resources (e.g., control channel elements (CCEs)) associated with encoded information for a control information format having a given payload size. Search space sets may include common search space sets configured for sending control information to multiple UEs **115** and UE-specific search space sets for sending control information to a specific UE **115**.

[0079] A network entity **105** may provide communication coverage via one or more cells, for example a macro cell, a small cell, a hot spot, or other types of cells, or any combination thereof. The term “cell” may refer to a logical communication entity used for communication with a network entity **105** (e.g., using a carrier) and may be associated with an identifier for distinguishing neighboring cells (e.g., a physical cell identifier (PCID), a virtual cell identifier (VCID), or others). In some examples, a cell also may refer to a coverage area **110** or a portion of a coverage area **110** (e.g., a sector) over which the logical communication entity operates. Such cells may range from smaller areas (e.g., a structure, a subset of structure) to larger areas depending on various factors such as the capabilities of the network entity **105**. For example, a cell may be or include a building, a subset of a building, or exterior spaces between or overlapping with coverage areas **110**, among other examples.

[0080] A macro cell generally covers a relatively large geographic area (e.g., several kilometers in radius) and may allow unrestricted access by the UEs **115** with service subscriptions with the network provider supporting the macro cell. A small cell may be associated with a lower-powered network entity **105** (e.g., a lower-powered base station **140**), as compared with a macro cell, and a small cell may operate using the same or different (e.g., licensed, unlicensed) frequency bands as macro cells. Small cells may provide unrestricted access to the UEs **115** with service subscriptions with the network provider or may provide restricted access to the UEs **115** having an association with the small cell (e.g., the UEs **115** in a closed subscriber group (CSG), the UEs **115** associated with users in a home or office). A network entity **105** may support one or multiple cells and may also support communications via the one or more cells using one or multiple component carriers.

[0081] In some examples, a carrier may support multiple cells, and different cells may be configured according to different protocol types (e.g., MTC, narrowband IoT (NB-IoT), enhanced mobile broadband (eMBB)) that may provide access for different types of devices.

[0082] In some examples, a network entity **105** (e.g., a base station **140**, an RU **170**) may be movable and therefore provide communication coverage for a moving coverage area **110**. In some examples, different coverage areas **110** associated with different technologies may overlap, but the different coverage areas **110** may be supported by the same network entity **105**. In some other examples, the overlapping coverage areas **110** associated with different technologies may be supported by different network entities **105**. The wireless communications system **100** may include, for example, a heterogeneous network in which different types of the network entities **105** provide coverage for various coverage areas **110** using the same or different radio access technologies.

[0083] Some UEs **115**, such as MTC or IoT devices, may be low cost or low complexity devices and may provide for automated communication between machines (e.g., via Machine-to-Machine (M2M) communication). M2M communication or MTC may refer to data communication technologies that allow devices to communicate with one another or a network entity **105** (e.g., a base station **140**) without human intervention. In some examples, M2M communication or MTC may include communications from devices that integrate sensors or meters to measure or capture information and relay such information to a central server or application program that uses the information or presents the information to humans interacting with the application program. Some UEs **115** may be designed to collect information or enable automated behavior of machines or other devices. Examples of applications for MTC devices include smart metering, inventory monitoring, water level monitoring, equipment monitoring, healthcare monitoring, wildlife monitoring, weather and geological event monitoring, fleet management and tracking, remote security sensing, physical access control, and transaction-based business charging.

[0084] The wireless communications system **100** may be configured to support ultra-reliable communications or low-latency communications, or various combinations thereof. For example, the wireless communications system **100** may be configured to support ultra-reliable low-latency communications (URLLC). The UEs **115** may be designed to support ultra-reliable, low-latency, or critical functions. Ultra-reliable communications may include private communication or group communication and may be supported by one or more services such as push-to-talk, video, or data. Support for ultra-reliable, low-latency functions may include prioritization of services, and such services may be used for public safety or general commercial applications. The terms ultra-reliable, low-latency, and ultra-reliable low-latency may be used interchangeably herein.

[0085] In some examples, a UE **115** may be configured to support communicating directly with other UEs **115** via a device-to-device (D2D) communication link **135** (e.g., in accordance with a peer-to-peer (P2P), D2D, or sidelink protocol). In some examples, one or more UEs **115** of a group that are performing D2D communications may be within the coverage area **110** of a network entity **105** (e.g., a base station **140**, an RU **170**), which may support aspects of such D2D communications being configured by (e.g., scheduled by) the network entity **105**. In some examples, one or more UEs **115** of such a group may be outside the coverage area **110** of a network entity **105** or may be otherwise unable to or not configured to receive transmissions from a network entity **105**. In some examples, groups of the UEs **115** communicating via D2D communications may support a one-to-many (1:M) system in which each UE **115** transmits to each of the other UEs **115** in the group. In some examples, a network entity **105** may facilitate the scheduling of resources for D2D communications. In some other examples, D2D communications may be carried out between the UEs **115** without an involvement of a network entity **105**.

[0086] In some systems, a D2D communication link **135** may be an example of a communication channel, such as a sidelink communication channel, between vehicles (e.g., UEs **115**). In some examples, vehicles may communicate using vehicle-to-everything (V2X) communications, vehicle-to-vehicle (V2V) communications, or some combination of these. A vehicle may signal information related to traffic conditions, signal scheduling, weather, safety, emergencies, or any other information relevant to a V2X system. In some examples, vehicles in a V2X system may communicate with roadside infrastructure, such as roadside units, or with the network via one or more network nodes (e.g., network entities **105**, base stations **140**, RUs **170**) using vehicle-to-network (V2N) communications, or with both.

[0087] The core network **130** may provide user authentication, access authorization, tracking, Internet Protocol (IP) connectivity, and other access, routing, or mobility functions. The core network **130** may be an evolved packet core (EPC) or 5G core (5GC), which may include at least one control plane entity that manages access and mobility (e.g., a mobility management entity (MME), an access and mobility management function (AMF)) and at least one user plane entity that routes packets or interconnects to external networks (e.g., a serving gateway (S-GW), a Packet Data Network (PDN) gateway (P-GW), or a user plane function (UPF)). The control plane entity may manage non-access stratum (NAS) functions such as mobility, authentication, and bearer management for the UEs **115** served by the network entities **105** (e.g., base stations **140**) associated with the core network **130**. User IP packets may be transferred through the user plane entity, which may provide IP address allocation as well as other functions. The user plane entity may be connected to IP services **150** for one or more network operators. The IP services **150** may include access to the Internet, Intranet(s), an IP Multimedia Subsystem (IMS), or a Packet-Switched Streaming Service.

[0088] The wireless communications system **100** may operate using one or more frequency bands, which may be in the range of 300 megahertz (MHz) to 300 gigahertz (GHz). Generally, the region from 300 MHz to 3 GHz is known as the ultra-high frequency (UHF) region or decimeter band because the wavelengths range from approximately one decimeter to one meter in length. UHF waves may be blocked or redirected by buildings and environmental features, which may be referred to as clusters, but the waves may penetrate structures sufficiently for a macro cell to provide service to the UEs **115** located indoors. Communications using UHF waves may be associated with smaller antennas and shorter ranges (e.g., less than 100 kilometers) compared to communications using the smaller frequencies and longer waves of the high frequency (HF) or very high frequency (VHF) portion of the spectrum below 300 MHz.

[0089] The wireless communications system **100** may utilize both licensed and unlicensed RF spectrum bands. For example, the wireless communications system **100** may employ License Assisted Access (LAA), LTE-Unlicensed (LTE-U) radio access technology, or NR technology using an unlicensed band such as the 5 GHz industrial, scientific, and medical (ISM) band. While operating using unlicensed RF spectrum bands, devices such as the network entities **105** and the UEs **115** may employ carrier sensing for collision detection and avoidance. In some examples, operations using unlicensed bands may be based on a carrier aggregation configuration in conjunction with component carriers operating using a licensed band (e.g., LAA). Operations using unlicensed spectrum may include downlink transmissions, uplink transmissions, P2P transmissions, or D2D transmissions, among other examples.

[0090] A network entity **105** (e.g., a base station **140**, an RU **170**) or a UE **115** may be equipped with multiple antennas, which may be used to employ techniques such as transmit diversity, receive diversity, multiple-input multiple-output (MIMO) communications, or beamforming. The antennas of a network entity **105** or a UE **115** may be located within one or more antenna arrays or antenna panels, which may support MIMO operations or transmit or receive beamforming. For example, one or more base station antennas or antenna arrays may be co-located at an antenna assembly, such as an antenna tower. In some examples, antennas or antenna arrays associated with a network entity **105** may be located at diverse geographic locations. A network entity **105** may include an antenna array with a set of rows and columns of antenna ports that the network entity **105** may use to support beamforming of communications with a UE **115**. Likewise, a UE **115** may include one or more antenna arrays that may support various MIMO or beamforming operations. Additionally, or alternatively, an antenna panel may support RF beamforming for a signal transmitted via an antenna port.

[0091] Beamforming, which may also be referred to as spatial filtering, directional transmission, or directional reception, is a signal processing technique that may be used at a transmitting device or a receiving device (e.g., a network entity **105**, a UE **115**) to shape or steer an antenna beam (e.g., a transmit beam, a receive beam) along a spatial path between the transmitting device and the receiving device. Beamforming may be achieved by combining the signals communicated via antenna elements of an antenna array such that some signals propagating along particular orientations with respect to an antenna array experience constructive interference while others experience destructive interference. The adjustment of signals communicated via the antenna elements may include a transmitting device or a receiving device applying amplitude offsets, phase offsets, or both to signals carried via the antenna elements associated with the device. The adjustments associated with each of the antenna elements may be defined by a beamforming weight set associated with a particular orientation (e.g., with respect to the antenna array of the transmitting device or receiving device, or with respect to some other orientation).

[0092] The wireless communications system **100** may be a packet-based network that operates according to a layered protocol stack. In the user plane, communications at the bearer or PDCP layer may be IP-based. An RLC layer may perform packet segmentation and reassembly to communicate via logical channels. A MAC layer may perform priority handling and multiplexing of logical channels into transport channels. The MAC layer also may implement error detection techniques, error correction techniques, or both to support retransmissions to improve link efficiency. In the control plane, an RRC layer may provide establishment, configuration, and maintenance of an RRC connection between a UE **115** and a network entity **105** or a core network **130** supporting radio bearers for user plane data. A PHY layer may map transport channels to physical channels.

[0093] In some examples, the wireless communications system **100** may support communications with aircrafts operating in the vicinity of network entities **105**, coverage areas **110**, and/or UEs **115**. Some aircraft may implement DAA systems to support collision or conflict avoidance, and these systems may rely on human observers, ground-based observations systems, remote human pilots, and/or on-board systems.

[0094] Techniques described herein support aircraft DAA systems using ground-based network infrastructure. For example, a server (e.g., a local DAA server) may be positioned in the wireless communications system **100** to support DAA systems at aircraft operating in the wireless communications system **100**. The server may be part of or operate in conjunction with a network entity **105** for communications with aircraft, which may implement or function as a UE **115**, as described herein. The server may receive information (e.g., via a radio access network (RAT)) from an aircraft that is operating in an autonomous flying mode. The information may be associated with the aircraft and may include general awareness of other aircraft, a potential conflict identified by the aircraft and/or metadata associated with the aircraft. The metadata may include an aircraft type, capability, position, velocity, identify, operator information, or a combination thereof.

[0095] In response to receiving the information for the aircraft, the server may transmit situational information to the aircraft. The situational information may include a warning of a potential conflict, an avoidance instructions, and/or neighboring vehicle information for vehicles operating in the zone or in another proximal zone. The server may identify such information in response to receiving the information from the aircraft. In some examples, the server may communicate information associated with an aircraft with a USS via a network function. For example, the server may retrieve publicly available information (e.g., flight plan, mission type, operator information) from the USS to support the DAA systems at the aircraft or the server. The server may also communicate with various sensors to support DAA systems. As such, techniques described herein may utilize aspects of the wireless communications systems **100** to support DAA techniques in aircraft, thereby improving aircraft safety and reliability.

[0096] FIG. **2** illustrates an example of a wireless communications system **200** that supports network assisted detect and avoid systems for aerial vehicles in accordance with one or more aspects of the present disclosure. The wireless communications system **200** includes aircrafts **205**, a server **210**, and a USS **215**. The aircrafts **205** may be examples of aircrafts that operate in an autonomous flying mode as described herein. For example, the aircraft **205** may be UAVs, drones, remote piloted aircrafts, piloted aircrafts, or the like. In the case of remote piloted or piloted aircrafts, the aircrafts **205** may operate in an autonomous mode, such as by using an autopilot feature.

[0097] The aircrafts **205** may be implemented with DAA systems (or sense and avoid (SAA) systems) which may allow UAVs and drones to integrate into civilian airspace by avoiding collisions with other aircrafts, buildings, power lines, birds, and other obstacles. These systems are configured to observe an environment surrounding the aircraft **205**, determine whether a collision is imminent, and generate a new flight path in order to avoid a collision. UAV DAA/SAA systems may combine data from a number of sensors, using sensor fusion algorithms, image recognition, and artificial intelligence to support collision avoidance. Data may be fed back to the drone on-board computer and/or drone flight controller, which can then decide on a evasive maneuver or flight path correction to avoid collision. In some jurisdictions, a reliable onboard DAA system may be a prerequisite for obtaining an authorization or waiver for flight operations, which may typically require human observers or ground-based observation systems in the vicinity of the flight path. DAA systems may thus support commercially viable beyond visual line of sight (BVLOS) drone operations that provide services such as inspection and cargo delivery beyond the visual range of the operator or observers.

[0098] Some DAA solutions may be active or passive solutions that use sensor data. Passive solutions may use electro-optical data (e.g., cameras), passive radar and acoustic data. Active solutions may use sonar, lidar, and/or active radar. Some solutions may also be communications based. For example, UAVs may use systems designed for manned aviation, such as traffic collision and avoidance systems (TCAS) and/or automatic dependent surveillance-broadcast (ADS-B) systems that periodically broadcast and receive identity data, position data, and other information. Some standards may define DAA requirements based on airborne collision avoidance systems (ACAS). For UAS, three versions of ACAS are defined or are in development: (1) ACAS-Xu for fixed winged UAS; (2) ACAX-Xr-ACAS Xu adapted for rotocraft; and (3) ACAS-xXu for small UAS.

[0099] Techniques described herein support DAA without or with limited reliance on a remote pilot station (RPS), a UAV controller, a ground control station (GCS), and/or a human pilot. The techniques described herein may support DAA for automation techniques in aerial vehicles (aircraft **205**) but without sole reliance on aerial vehicle awareness of surrounding traffic using onboard sensor information. The techniques described herein may also assume U2U communications to collect traffic awareness information (e.g., ACAS sXu). These techniques are supported by using ground-based network for higher spatial awareness of traffic and other conditions.

[0100] To support ground-based awareness, the server **210** may be implemented in the wireless communications system **200** as a localized USS/UTM node configured for DAA. The server **210** may be implemented in RAN **220**. For example, the server **210** may be configured as or implemented with a network entity **105**, as described with respect to FIG. **1**. The server **210** may support artificial intelligence and/or machine learning techniques for traffic and/or condition awareness and forecasting at the aircraft **205**. In some examples, the server **210** provides a subscription-based traffic separation service, such that various aircraft **205** may subscribe to the services provided by the server **210**. The server **210** may be configured to support spatial awareness based on information collected from various vehicles (e.g., aircraft **205**), such as UAVs, and from other devices such as network entities **105** and other sources of information. The server **210** may implement traffic separation algorithms and collision notification features across one or more cells. A particular aircraft **205** may be visible to multiple servers **210**.

[0101] Additionally, to support DAA systems at the aircraft **205**, the server **210** may interact with and leverage a network data analytics function (NWDAF) and a network exposure function (NEF) (e.g., unmanned arial systems network function (UAS NF) **225**) to interact with unmanned aircraft system traffic management (UTM) systems and USSs (e.g., USS **215**). The NWDAF and the NEF may provide access to network services and capabilities via a network (e.g., a 5G core network **230**). The 5G core network **230** may be an example of core network **130** of FIG. **1**. As described in further detail herein, the server **210** may provide, via a NEF (e.g., UAS NF **225**), aerial congestion information and aircraft **205** information to the USS **215** to support in flight authorization.

[0102] USSs (e.g., USS **215**) may offer services to support efficient and safe utilization of civilian airspace by unmanned aerial systems (e.g., aircraft **205**). The USS **215** may be an example of or used to support IP services **150** of FIG. **1**. The USS **215** be an example of aspects of unmanned aircraft system traffic management infrastructure (UTMI) **235**. The USS **215** may function as communication endpoint between aircraft (e.g., aircraft **205**) and authorities. The USS may support tools for traffic management, flight planning, operational data, and the like. Aspects of the UTMI

235, such as the USS **215**, may be accessible by RAN **220** services (e.g., the server **210**) via a NEF, such as the UAS NF **225**.

[0103] As an aircraft, such as aircraft **205-***a,* enters a zone or airspace associated with the server **210**, then the communications between the aircraft **205-***a* and the server **210** may be initiated by the aircraft **205-***a* and or server **210**. For example, the aircraft **205-***a* may transmit first information associated with the aircraft **205-***a* to the server **210** via the RAN **220**. The first information may include metadata associated with the aircraft **205-***a,* such as an aircraft type, aircraft capability, an aircraft position, an aircraft velocity, an aircraft identify, operator information, or a combination thereof. The aircraft identity may be an example of a broadcast remote identifier (BRID) associated with the aircraft **205-***a.* In some examples, the first information associated with the aircraft **205-***a* that is transmitted to the server **210** may include neighboring vehicle information detected by the aircraft **205-***a.* For example, the aircraft **205-***a* may detect that aircraft **205-***b* and **205-***c* are operating nearby and indicate such information to the server **210**. In some examples, the aircraft **205-***a* may receive identifiers (e.g., BRIDs) broadcast by the aircrafts **205-***b* and **205-***c* and transmit the identifiers to the server **210**.

[0104] In some examples, the aircraft **205-***a* may detect a possible conflict with another aircraft **205** or another object such as a powerline, building, ground vehicle, etc. In such cases, the aircraft **205-***a* may transmit a deconfliction request or a conflict warning to the server **210**. The deconfliction request or conflict warning may be associated with the potential conflict detected by the aircraft **205-***a.*

[0105] In response to receiving the first information associated with the aircraft **205-***a* and/or the deconfliction request or conflict warning, the server **210** may transmit situational information to the aircraft **205-***a.* The situational information may trigger a situational response (e.g., an evasive maneuver) by the aircraft **205-***a.* In some examples, the situation information may include an indication of an avoidance instruction for avoiding a conflict by the aircraft **205-***a.* For example, the avoidance instructions may include instructions for performing the evasive maneuver (e.g., ascend, descend, or turn). The situational information may also include a warning that is indicative of a potential conflict associated with the aircraft **205-***a.* The indication may include location information associated with the potential conflict. In such cases, the aircraft **205-***a* may determine the evasive maneuver to avoid the conflict. In some examples, the situational information includes neighboring vehicle information associated with vehicles operating in the zone (or another proximal zone) associated with the server **210**.

[0106] As described herein, the server **210** may communicate with the USS **215** via a network function (e.g., UAS NF **225**). The communication between the server **210** and the USS **215** may be performed in response to receiving the first information from the aircraft **205-***a.* In some cases, the server **210** and the USS **215** may communicate information associated with the aircraft **205-***a.* For example, the server **210** may transmit information associated with the aircraft, such an identifier of the aircraft **205-***a,* to the USS **215**. In some examples, the server **210** may transmit congestion or conflict information for the zone associated with the server **210** to the USS **215** to support flight planning at the USS **215**. For example, if the quantity of aircrafts in the zone associated with the server **210** is above—or anticipated to be above—a threshold, then the server **210** may transmit, to the USS **215**, an indication of the number of aircrafts and/or an indication that the number of aircrafts is above or anticipated to be above the threshold. Additionally, or alternatively, the server **210** may request information associated with the aircraft from the USS **215**. For example, the server **210** transmits an API request that includes an aircraft identifier associated with the aircraft **205-***a.* In response, the USS **215** may return information such as aircraft type or category, aircraft capability, operator information, mission type, or the like. This information may be used by the server **210** to determine spatial awareness information, situational information, or the like to be used by aircraft **205** in the area or zone associated with the server **210**.

[0107] The server **210** may also use sensor data received from sensors **240** to support DAA techniques at aircraft **205**-*a*. The sensors **240** may be collocated with the server **210** and/or positioned in a zone associated with the server **210**. For example, a aircraft identity sensor, such as a BRID receiver, may be positioned in the zone and may receive or detect aircraft identifiers that are broadcast by the aircraft **205**. The identifiers may be communicated to the server **210** to support DAA solutions as described herein. Thus, using the sensors **240** as BRID receivers, the server **210** may be able to identify aircraft **205** that are operating in the zone associated with the server **210**. In some examples, identity broadcasts may be limited by a coverage area, such as BRID coverage areas **245**. As such, positioning of sensors **240**-*a* in the zone associated with the server **210** may support advanced aircraft identification. The sensors **240** may be positioned in conjunction with network entities (e.g., network entities **105** of FIG. **1**) to support communications between the sensors **240** and the server **210**. The sensors may also be DAA broadcast receivers, automatic dependent surveillance-broadcast (ADS-B) receivers, weather sensors, radar, NR sensors, lidar, or the like. For example, a weather sensor may sense wind data which may support DAA solutions. In some examples, the sensor data received from the sensors **240** may be communicated to the USS **215** (e.g., via UAS NF **225**) to support flight planning, traffic planning and other solutions.

[0108] In one example use of the system described herein, the aircraft **205**-*a* may detect a possible conflict situation (e.g., based on on-board sensors), and the aircraft **205**-*a* may transmit an indication of the detected conflict to the server **210**. In response to receiving the indication of the detected conflict, the server **210**-*a* may collect data from the sensor **240**-*a*. Using the collected data and the indication received from the aircraft **205**-*a*, the server **210** may detect or identify a possible conflict situation. The server **210** may transmit a trigger warning to the aircraft **205**-*b* and an emergency directive (e.g., evasive maneuver instructions) to the aircraft **205**-*c*, which may trigger an evasive maneuver by the aircraft **205**-*c*. As such, the server **210** is used to support DAA techniques at multiple aircraft **205** operating in a zone associated with the server.

[0109] FIG. **3** illustrates an example of an operating environment **300** that supports network assisted detect and avoid systems for aerial vehicles in accordance with one or more aspects of the present disclosure. The operating environment includes an aircraft **305**, a DAA USS **315**, and a set of servers **310**. The aircraft **305** may be an example of an aircraft **205** as described with respect to FIG. **2**. The DAA USS **315** may be an example of a USS **315** as described with respect to FIG. **2**. The operating environment **300** also includes a discovery and synchronizations service (DSS) **320** and a DAA service area **325** that includes a set of servers **310**, which may be examples of the server **210** of FIG. **2**. Each server **310** may support a different and/or overlapping portion of the DAA service area **325**.

[0110] Some techniques support a DSS being used for USSs to discover each other in order to push/subscribe to entities of information that the respective USSs may provide. Techniques described herein leverage the DSS to support DAA service area discovery provisioning of such information for aircrafts. For example, a home USS **330** of the aircraft **305** may provide update updates on the DAA service area **325** topology and reachability (e.g., referred to as the DAA service area entity). Once the aircraft **305** has and maintains updates to the DAA service area entity, the aircraft **305** may have the geographic coverage areas (e.g., zones) of the current server **310** and reachability information (e.g., resource address, such as a URL) to support the DAA techniques described herein.

[0111] FIG. **3** illustrates example communications that are used to support DAA for aircraft **305** in the DAA service area **325**. At **335**, the aircraft **305**, or a control station associated with the aircraft **305**, may initiate and maintain a subscription to the local detect and avoid service via the home USS **330**. Thus, the aircraft **305** may obtain and maintain data associated with DAA service areas and their local zones and respective servers **310**, and the data may include and identification information to reach the local servers, such as uniform resource locator (URL) addresses. At **340**, the DAA USS may register the DAA service to the DSS **320**, and the DSS **320** may be configured with data associated with the DAA service area geography for reporting. At **345**, the home USS

330 may discover and subscribe to the DAA service area **325** description entity (e.g., an instance of discoverable information). The description entity may include information associated with the DAA service area **325** as well as local zones (e.g., a zone **350**). After subscribing to the DAA USS **315**, the home USS **330** may receive the entity information and or any updates to such information from the DAA USS **315** via a direct protocol at **355** (e.g., without the use of the DSS **320**).

[0112] With the information associated with the DAA service area **325** topology, the aircraft may initiate contact with and communicate with the servers **310**. After initiation of communication, the communication protocol may support server **310** discovery such that the aircraft **305** may not have to continue subscribing via the home USS **330**. In some examples, the aircraft **305** may receive information about another USS via an already subscribed USS (e.g., DAA USS **315**), thereby bypassing the need for receiving DAA service information directly from the home USS **330**.

[0113] As described herein, the DAA USS **315** may be a USS that manages one or more DAA service areas (e.g., service area **325**), each subdivided into LDS local zones (e.g., a zone **350**). Management of the DAA service area **325** may involve maintaining a database (e.g., local or remote) of zone descriptors (e.g., server **310** identifying information and location information), such as server identifiers and/or resource addresses (e.g., layer 2, IP, URI or application layer). A zone may be an example of a local zone that is supported by a server **310** for DAA techniques. More than one server **310** may be assigned to a zone. As described herein, the DSS **320** may be used to discover and maintain awareness of geographically-defined services and information sets (e.g., USSs). The home USS **330** may be used by the aircraft **305** to obtain the database of service area descriptions and may be used to maintain updates to the database. The aircraft **305** may communicate with the home USS **330** over a LDS discovery and subscription service protocol, which may be a proprietary protocol. The home USS **330** may also support an associated DAA service area. The servers **310** may be assigned/configured to a zone by a managing USS, such as USS **315**.

[0114] FIG. **4** illustrates an example of a process flow **400** that supports network assisted detect and avoid systems for aerial vehicles in accordance with one or more aspects of the present disclosure. The process flow **400** includes an aircraft **405**, a server **410**, a USS **415**, and a home USS **320**, which may be an example of the corresponding devices as described with respect to FIGS. **1** through **3**. The aircraft **405** may be subscribed to the home USS **420** in order to receive DAA entity information, among other information, to support DAA techniques as described herein. In some examples, some signaling or procedure of the process flow **400** may occur in different orders than shown. Additionally, or alternatively, some additional procedures of signaling may occur, or some signaling or procedures may not occur.

[0115] At **425**, the home USS **420** may receive, from a DSS, information associated with a second USS (e.g., USS **415**). The home USS **420** may receive the information in response to subscribing to the DSS. In some examples, receiving, the information for the second USS may include receiving and endpoint or identifier associated with the second USS.

[0116] At **430**, the home USS **420** may receive, from the second USS **415** based at least in part on receiving the information associated with the second USS **415**, detect and avoid service area entity information associated with one or more detect and avoid servers (e.g., server **410**) managed by the second USS. In some examples, the home USS **420** may transmit, to the second USS **415** in response to receiving the information, a request for the detect and avoid service area entity information, and the detect and avoid service area entity information is received based at least in part on transmitting the request. The detect and avoid service area entity information may include an indication of an reachability information associated with the server (e.g., an address), such as a uniform resource locator (URL) address, a uniform resource identifier (URI), an IP address, or another endpoint identifier for the one or more detected and avoid servers, a location of the one or more detect avoid servers, a zone identifier associated with the one or more detect and avoid servers, or a combination thereof.

[0117] At **435**, the home USS **320** may transmit, and the aircraft **405** may receive, an indication of the detect and avoid service area entity information that the aircraft is use while operating in an autonomous flying mode.

[0118] At **440**, the aircraft **405** may receive first information associated with one or more second vehicles. The aircraft **405** is operating in an autonomous flying mode. The first information may be detected by the aircraft **405**. In some examples, the first information may be aircraft identifiers associated with aircraft (e.g., second vehicles). In some examples, the second vehicles may be ground vehicles, other aircraft, piloted aircraft, and/or remote piloted aircraft.

[0119] At **445**, the aircraft **405** may transmit, and the server **410** may receive, via a radio access network and from the aircraft **405** operating in an autonomous flying mode, information associated with the aircraft **405**. The information may be an example of a deconfliction request or conflict warning associated with a potential conflict detected by the aircraft **405**. The information may be metadata associated with the aircraft **405**, and the metadata may include an aircraft type, an aircraft capability, an aircraft position, an aircraft velocity, an aircraft identity, aircraft operator information, or a combination thereof. The information may include information associated with other vehicles (e.g., neighboring aircraft) detected by the aircraft **405**. In some examples, the server **410** may determine, based at least in part on receiving the first information, that a potential conflict exists for the aircraft, a second aircraft, or both.

[0120] At **450**, the server **410** may transmit, and the aircraft **405** may receive, situational information to trigger a situational response by the aircraft **405** while the aircraft **405** operates in a zone associated with the server **410**. The situational information may include an indication of a potential conflict associated with the aircraft **405**, an indication of an avoidance instruction for avoiding a potential conflict by the aircraft **405**, and/or neighboring vehicle information associated with one or more vehicles operating in the zone. The situational information may additionally or alternatively include topographical information associated with the zone, ground-based obstruction information associated with the zone, or both. For example, the information may indication locations of topographical obstructions (e.g., mountains) and/or obstructions (e.g., buildings or power lines). The situational information may also include information associated with proximal zones, such as zones that are adjacent to the zone in which the aircraft **405** is operating, zones that are nearby the aircraft, etc. The server **410** and the aircraft **405** may communicate via an access link (e.g., a Uu link), a sidelink (e.g., a PC5 interface), or both. The server **410** and the aircraft **405** may communicate via layer 2 (L2) or layer 3 (L3) protocols. In examples when the aircraft **405** receives avoidance instructions (e.g., deconfliction advice), the aircraft **405** may not execute on the instructions without input from a pilot (e.g., remote pilot or on-board pilot) or a ground station/controller.

[0121] At **455**, the aircraft **405** may perform an evasive maneuver in response to receiving the situational information in order to avoid the aircraft. As described herein, the aircraft **405** may perform an evasive maneuver in response to receiving the situational information and input from a pilot or ground station/controller to perform the evasive maneuver.

[0122] At **460**, the server **410** may communicate between the server **410** and the USS **415**, second information associated with the aircraft **405**. The communicating may be performed via a network function based at least in part on receiving the first information associated with the aircraft **405**. For example, the server **410** may transmit, to the USS **415**, an indication of conflict information associated with the aircraft, an indication of vehicle congestion information of the zone, or a combination thereof. The server **410** may also transmit, to the USS **415**, a request for information associated with the aircraft **405**, such as aircraft type and/or operator information that may be used to determine situational information to transmit to the aircraft **405**. In some examples, the server **410** may receive from one or more sensors associated with the server, zone information associated with the zone, wherein the situational information is based at least in part on the zone information. The zone information may include other aircraft or vehicles detected by the sensors, weather information, ADS-B information, radar information, lidar information, or the like. The zone information may additionally include static or ephemeral ground-based obstacles such as structures, cranes, or topographical obstructions.

[0123] FIG. **5** shows a block diagram **500** of a device **505** that supports network assisted detect and avoid systems for aerial vehicles in accordance with one or more aspects of the present disclosure. The device **505** may be an example of aspects of a network entity **105** as described herein. The device **505** may include a receiver **510**, a transmitter **515**, and a communications manager **520**. The device **505** may also include a processor. Each of these components may be in communication with one another (e.g., via one or more buses).

[0124] The receiver **510** may provide a means for obtaining (e.g., receiving, determining, identifying) information such as user data, control information, or any combination thereof (e.g., I/Q samples, symbols, packets, protocol data units, service data units) associated with various channels (e.g., control channels, data channels, information channels, channels associated with a protocol stack). Information may be passed on to other components of the device **505**. In some examples, the receiver **510** may support obtaining information by receiving signals via one or more antennas. Additionally, or alternatively, the receiver **510** may support obtaining information by receiving signals via one or more wired (e.g., electrical, fiber optic) interfaces, wireless interfaces, or any combination thereof.

[0125] The transmitter **515** may provide a means for outputting (e.g., transmitting, providing, conveying, sending) information generated by other components of the device **505**. For example, the transmitter **515** may output information such as user data, control information, or any combination thereof (e.g., I/Q samples, symbols, packets, protocol data units, service data units) associated with various channels (e.g., control channels, data channels, information channels, channels associated with a protocol stack). In some examples, the transmitter **515** may support outputting information by transmitting signals via one or more antennas. Additionally, or alternatively, the transmitter **515** may support outputting information by transmitting signals via one or more wired (e.g., electrical, fiber optic) interfaces, wireless interfaces, or any combination thereof. In some examples, the transmitter **515** and the receiver **510** may be co-located in a transceiver, which may include or be coupled with a modem.

[0126] The communications manager **520**, the receiver **510**, the transmitter **515**, or various combinations thereof or various components thereof may be examples of means for performing various aspects of network assisted detect and avoid systems for aerial vehicles as described herein. For example, the communications manager **520**, the receiver **510**, the transmitter **515**, or various combinations or components thereof may support a method for performing one or more of the functions described herein.

[0127] In some examples, the communications manager **520**, the receiver **510**, the transmitter **515**, or various combinations or components thereof may be implemented in hardware (e.g., in communications management circuitry). The hardware may include a processor, a DSP, a CPU, an ASIC, an FPGA or other programmable logic device, a microcontroller, discrete gate or transistor logic, discrete hardware components, or any combination thereof configured as or otherwise supporting a means for performing the functions described in the present disclosure. In some examples, a processor and memory coupled with the processor may be configured to perform one or more of the functions described herein (e.g., by executing, by the processor, instructions stored in the memory).

[0128] Additionally, or alternatively, in some examples, the communications manager **520**, the receiver **510**, the transmitter **515**, or various combinations or components thereof may be implemented in code (e.g., as communications management software or firmware) executed by a processor. If implemented in code executed by a processor, the functions of the communications manager **520**, the receiver **510**, the transmitter **515**, or various combinations or components thereof may be performed by a general-purpose processor, a DSP, a CPU, an ASIC, an FPGA, a microcontroller, or any combination of these or other programmable logic devices (e.g., configured as or otherwise supporting a means for performing the functions described in the present disclosure).

[0129] In some examples, the communications manager **520** may be configured to perform various operations (e.g., receiving, obtaining, monitoring, outputting, transmitting) using or otherwise in cooperation with the receiver **510**, the transmitter **515**, or both. For example, the communications manager **520** may receive information from the receiver **510**, send information to the transmitter **515**, or be integrated in combination with the receiver **510**, the transmitter **515**, or both to obtain information, output information, or perform various other operations as described herein.

[0130] The communications manager **520** may support wireless communications at a server in accordance with examples as disclosed herein. For example, the communications manager **520** may be configured as or otherwise support a means for receiving, via a radio access network and from an aircraft, first information associated with the aircraft. The communications manager **520** may be configured as or otherwise support a means for transmitting, to the aircraft in response to receiving the first information, situational information to trigger a situational response by the aircraft while the aircraft operates in a zone associated with the server. The communications manager **520** may be configured as or otherwise support a means for communicating, between the server and an unmanned aircraft service supplier, second information associated with the aircraft, the communicating being via a network function and based on receiving the first information.

[0131] Additionally, or alternatively, the communications manager **520** may support communications at a first unmanned aircraft service supplier in accordance with examples as disclosed herein. For example, the communications manager **520** may be configured as or otherwise support a means for receiving, from a discovery and synchronization service, information associated with a second unmanned aircraft service supplier. The communications manager **520** may be configured as or otherwise support a means for receiving, from the second unmanned aircraft service supplier based on receiving the information, detect and avoid service area entity information associated with one or more detect and avoid servers managed by the second unmanned aircraft service supplier. The communications manager **520** may be configured as or otherwise support a means for transmitting, to an aircraft, an indication of the detect and avoid service area entity information that the aircraft is use while operating in an autonomous flying mode.

[0132] By including or configuring the communications manager **520** in accordance with examples as described herein, the device **505** (e.g., a processor controlling or otherwise coupled with the receiver **510**, the transmitter **515**, the communications manager **520**, or a combination thereof) may support techniques for improved aircraft safety and reliability by leveraging wireless communication networks to support DAA techniques.

[0133] FIG. **6** shows a block diagram **600** of a device **605** that supports network assisted detect and avoid systems for aerial vehicles in accordance with one or more aspects of the present disclosure. The device **605** may be an example of aspects of a device **505** or a network entity **105** as described herein. The device **605** may include a receiver **610**, a transmitter **615**, and a communications manager **620**. The device **605** may also include a processor. Each of these components may be in communication with one another (e.g., via one or more buses).

[0134] The receiver **610** may provide a means for obtaining (e.g., receiving, determining, identifying) information such as user data, control information, or any combination thereof (e.g., I/Q samples, symbols, packets, protocol data units, service data units) associated with various channels (e.g., control channels, data channels, information channels, channels associated with a protocol stack). Information may be passed on to other components of the device **605**. In some examples, the receiver **610** may support obtaining information by receiving signals via one or more antennas. Additionally, or alternatively, the receiver **610** may support obtaining information by receiving signals via one or more wired (e.g., electrical, fiber optic) interfaces, wireless interfaces, or any combination thereof.

[0135] The transmitter **615** may provide a means for outputting (e.g., transmitting, providing, conveying, sending) information generated by other components of the device **605**. For example, the transmitter **615** may output information such as user data, control information, or any combination thereof (e.g., I/Q samples, symbols, packets, protocol data units, service data units) associated with various channels (e.g., control channels, data channels, information channels, channels associated with a protocol stack). In some examples, the transmitter **615** may support outputting information by transmitting signals via one or more antennas. Additionally, or alternatively, the transmitter **615** may support outputting information by transmitting signals via one or more wired (e.g., electrical, fiber optic) interfaces, wireless interfaces, or any combination thereof. In some examples, the transmitter **615** and the receiver **610** may be co-located in a transceiver, which may include or be coupled with a modem.

[0136] The device **605**, or various components thereof, may be an example of means for performing various aspects of network assisted detect and avoid systems for aerial vehicles as described herein. For example, the communications manager **620** may include an aircraft information component **625**, a situational information component **630**, an USS interface **635**, a DSS interface **640**, an aircraft interface **645**, or any combination thereof. The communications manager **620** may be an example of aspects of a communications manager **520** as described herein. In some examples, the communications manager **620**, or various components thereof, may be configured to perform various operations (e.g., receiving, obtaining, monitoring, outputting, transmitting) using or otherwise in cooperation with the receiver **610**, the transmitter **615**, or both. For example, the communications manager **620** may receive information from the receiver **610**, send information to the transmitter **615**, or be integrated in combination with the receiver **610**, the transmitter **615**, or both to obtain information, output information, or perform various other operations as described herein.

[0137] The communications manager **620** may support wireless communications at a server in accordance with examples as disclosed herein. The aircraft information component **625** may be configured as or otherwise support a means for receiving, via a radio access network and from an aircraft, first information associated with the aircraft. The situational information component **630** may be configured as or otherwise support a means for transmitting, to the aircraft in response to receiving the first information, situational information to trigger a situational response by the aircraft while the aircraft operates in a zone associated with the server. The USS interface **635** may be configured as or otherwise support a means for communicating, between the server and an unmanned aircraft service supplier, second information associated with the aircraft, the communicating being via a network function and based on receiving the first information.

[0138] Additionally, or alternatively, the communications manager **620** may support communications at a first unmanned aircraft service supplier in accordance with examples as disclosed herein. The DSS interface **640** may be configured as or otherwise support a means for receiving, from a discovery and synchronization service, information associated with a second unmanned aircraft service supplier. The USS interface **635** may be configured as or otherwise support a means for receiving, from the second unmanned aircraft service supplier based on receiving the information, detect and avoid service area entity information associated with one or more detect and avoid servers managed by the second unmanned aircraft service supplier. The aircraft interface **645** may be configured as or otherwise support a means for transmitting, to an aircraft, an indication of the detect and avoid service area entity information that the aircraft can use while operating in an autonomous flying mode.

[0139] FIG. **7** shows a block diagram **700** of a communications manager **720** that supports network assisted detect and avoid systems for aerial vehicles in accordance with one or more aspects of the present disclosure. The communications manager **720** may be an example of aspects of a communications manager **520**, a communications manager **620**, or both, as described herein. The communications manager **720**, or various components thereof, may be an example of means for performing various aspects of network assisted detect and avoid systems for aerial vehicles as described herein. For example, the communications manager **720** may include an aircraft information component **725**, a situational information component **730**, an USS interface **735**, a DSS interface **740**, an aircraft interface **745**, an avoidance instruction component **750**, a neighboring vehicle component **755**, a request or warning interface **760**, a metadata component **765**, a conflict component **770**, a sensor interface **775**, or any combination thereof. Each of these components may communicate, directly or indirectly, with one another (e.g., via one or more buses) which may include communications within a protocol layer of a protocol stack, communications associated with a logical channel of a protocol stack (e.g., between protocol layers of a protocol stack, within a device, component, or virtualized component associated with a network entity **105**, between devices, components, or virtualized components associated with a network entity **105**), or any combination thereof.

[0140] The communications manager **720** may support wireless communications at a server in accordance with examples as disclosed herein. The aircraft information component **725** may be configured as or otherwise support a means for receiving, via a radio access network and from an aircraft, first information associated with the aircraft. The situational information component **730** may be configured as or otherwise support a means for transmitting, to the aircraft in response to receiving the first information, situational information to trigger a situational response by the aircraft while the aircraft operates in a zone associated with the server. The USS interface **735** may be configured as or otherwise support a means for communicating, between the server and an unmanned aircraft service supplier, second information associated with the aircraft, the communicating being via a network function and based on receiving the first information.

[0141] In some examples, to support transmitting the situational information, the situational information component **730** may be configured as or otherwise support a means for transmitting a warning indicative of a potential conflict associated with the aircraft.

[0142] In some examples, to support transmitting the situational information, the avoidance instruction component **750** may be configured as or otherwise support a means for transmitting an indication of an avoidance instruction for avoiding a potential conflict by the aircraft.

[0143] In some examples, to support transmitting the situational information, the neighboring vehicle component **755** may be configured as or otherwise support a means for transmitting neighboring vehicle information associated with one or more vehicles operating in the zone or another proximal zone.

[0144] In some examples, to support transmitting the situational information, the neighboring vehicle component **755** may be configured as or otherwise support a means for transmitting topographical information, ground-based obstruction information, or both associated with the zone or another proximal zone.

[0145] In some examples, to support receiving the first information, the request or warning interface **760** may be configured as or otherwise support a means for receiving a deconfliction request or conflict warning associated with a potential conflict detected by the aircraft, where situational information is transmitted based on receiving the deconfliction request or the conflict warning and includes an avoidance instruction.

[0146] In some examples, to support receiving the first information, the metadata component **765** may be configured as or otherwise support a means for receiving metadata associated with the aircraft, where the metadata includes an aircraft type, an aircraft capability, an aircraft position, an aircraft velocity, an aircraft identity, aircraft operator information, or a combination thereof.

[0147] In some examples, the aircraft information component **725** may be configured as or otherwise support a means for receiving, from the aircraft, third information associated with one or more vehicles detected by the aircraft.

[0148] In some examples, the one or more vehicles include a piloted aircraft, an aircraft operating

in the autonomous flying mode, a ground vehicle, or a combination thereof.
[0149] In some examples, to support communicating the second information, the USS interface **735** may be configured as or otherwise support a means for transmitting, to the unmanned aircraft service supplier, an indication of conflict information associated with the aircraft, an indication of vehicle congestion information of the zone, or a combination thereof.
[0150] In some examples, to support communicating the second information, the USS interface **735** may be configured as or otherwise support a means for transmitting, to the unmanned aircraft service supplier, a request for the second information associated with the aircraft. In some examples, to support communicating the second information, the USS interface **735** may be configured as or otherwise support a means for receiving, from the unmanned aircraft service supplier, in response to transmitting the request, the second information.
[0151] In some examples, the conflict component **770** may be configured as or otherwise support a means for determining, based on receiving the first information, that a potential conflict exists for the aircraft, a second aircraft, or both. In some examples, the situational information component **730** may be configured as or otherwise support a means for transmitting, to the aircraft, the second aircraft, or both, an indication of the potential conflict.
[0152] In some examples, the sensor interface **775** may be configured as or otherwise support a means for receiving, from one or more sensors associated with the server, zone information associated with the zone, where the situational information is based on the zone information.
[0153] In some examples, the one or more sensors include a detect and avoid broadcast receiver, a broadcast remote identifier receiver, an automatic dependent surveillance-broadcast receiver, a weather sensor, a radar, a new-radio sensor, a lidar, an aircraft transponder, or a combination thereof.
[0154] In some examples, the server communicates with the aircraft via an access link, a sidelink, or both.
[0155] Additionally, or alternatively, the communications manager **720** may support communications at a first unmanned aircraft service supplier in accordance with examples as disclosed herein. The DSS interface **740** may be configured as or otherwise support a means for receiving, from a discovery and synchronization service, information associated with a second unmanned aircraft service supplier. In some examples, the USS interface **735** may be configured as or otherwise support a means for receiving, from the second unmanned aircraft service supplier based on receiving the information, detect and avoid service area entity information associated with one or more detect and avoid servers managed by the second unmanned aircraft service supplier. The aircraft interface **745** may be configured as or otherwise support a means for transmitting, to an aircraft, an indication of the detect and avoid service area entity information that the aircraft can use while operating in an autonomous flying mode.
[0156] In some examples, to support receiving the information associated with the second unmanned aircraft service supplier, the DSS interface **740** may be configured as or otherwise support a means for receiving an indication of reachability information for the one or more detected and avoid servers, a location of the one or more detect avoid servers, a zone identifier associated with the one or more detect and avoid servers, or a combination thereof.
[0157] In some examples, the USS interface **735** may be configured as or otherwise support a means for transmitting, to the second unmanned aircraft service supplier in response to receiving the information, a request for the detect and avoid service area entity information, where the detect and avoid service area entity information is received based on transmitting the request.
[0158] FIG. **8** shows a diagram of a system **800** including a device **805** that supports network assisted detect and avoid systems for aerial vehicles in accordance with one or more aspects of the present disclosure. The device **805** may be an example of or include the components of a device **505**, a device **605**, or a network entity **105** as described herein. The device **805** may communicate with one or more network entities **105**, one or more UEs **115**, or any combination thereof, which

may include communications over one or more wired interfaces, over one or more wireless interfaces, or any combination thereof. The device **805** may include components that support outputting and obtaining communications, such as a communications manager **820**, a transceiver **810**, an antenna **815**, a memory **825**, code **830**, and a processor **835**. These components may be in electronic communication or otherwise coupled (e.g., operatively, communicatively, functionally, electronically, electrically) via one or more buses (e.g., a bus **840**).

[0159] The transceiver **810** may support bi-directional communications via wired links, wireless links, or both as described herein. In some examples, the transceiver **810** may include a wired transceiver and may communicate bi-directionally with another wired transceiver. Additionally, or alternatively, in some examples, the transceiver **810** may include a wireless transceiver and may communicate bi-directionally with another wireless transceiver. In some examples, the device **805** may include one or more antennas **815**, which may be capable of transmitting or receiving wireless transmissions (e.g., concurrently). The transceiver **810** may also include a modem to modulate signals, to provide the modulated signals for transmission (e.g., by one or more antennas **815**, by a wired transmitter), to receive modulated signals (e.g., from one or more antennas **815**, from a wired receiver), and to demodulate signals. The transceiver **810**, or the transceiver **810** and one or more antennas **815** or wired interfaces, where applicable, may be an example of a transmitter **515**, a transmitter **615**, a receiver **510**, a receiver **610**, or any combination thereof or component thereof, as described herein. In some examples, the transceiver may be operable to support communications via one or more communications links (e.g., a communication link **125**, a backhaul communication link **120**, a midhaul communication link **162**, a fronthaul communication link **168**).

[0160] The memory **825** may include RAM and ROM. The memory **825** may store computer-readable, computer-executable code **830** including instructions that, when executed by the processor **835**, cause the device **805** to perform various functions described herein. The code **830** may be stored in a non-transitory computer-readable medium such as system memory or another type of memory. In some cases, the code **830** may not be directly executable by the processor **835** but may cause a computer (e.g., when compiled and executed) to perform functions described herein. In some cases, the memory **825** may contain, among other things, a BIOS which may control basic hardware or software operation such as the interaction with peripheral components or devices.

[0161] The processor **835** may include an intelligent hardware device (e.g., a general-purpose processor, a DSP, an ASIC, a CPU, an FPGA, a microcontroller, a programmable logic device, discrete gate or transistor logic, a discrete hardware component, or any combination thereof). In some cases, the processor **835** may be configured to operate a memory array using a memory controller. In some other cases, a memory controller may be integrated into the processor **835**. The processor **835** may be configured to execute computer-readable instructions stored in a memory (e.g., the memory **825**) to cause the device **805** to perform various functions (e.g., functions or tasks supporting network assisted detect and avoid systems for aerial vehicles). For example, the device **805** or a component of the device **805** may include a processor **835** and memory **825** coupled with the processor **835**, the processor **835** and memory **825** configured to perform various functions described herein. The processor **835** may be an example of a cloud-computing platform (e.g., one or more physical nodes and supporting software such as operating systems, virtual machines, or container instances) that may host the functions (e.g., by executing code **830**) to perform the functions of the device **805**.

[0162] In some examples, a bus **840** may support communications of (e.g., within) a protocol layer of a protocol stack. In some examples, a bus **840** may support communications associated with a logical channel of a protocol stack (e.g., between protocol layers of a protocol stack), which may include communications performed within a component of the device **805**, or between different components of the device **805** that may be co-located or located in different locations (e.g., where the device **805** may refer to a system in which one or more of the communications manager **820**, the transceiver **810**, the memory **825**, the code **830**, and the processor **835** may be located in one of the different components or divided between different components).

[0163] In some examples, the communications manager **820** may manage aspects of communications with a core network **130** (e.g., via one or more wired or wireless backhaul links). For example, the communications manager **820** may manage the transfer of data communications for client devices, such as one or more UEs **115**. In some examples, the communications manager **820** may manage communications with other network entities **105**, and may include a controller or scheduler for controlling communications with UEs **115** in cooperation with other network entities **105**. In some examples, the communications manager **820** may support an X2 interface within an LTE/LTE-A wireless communications network technology to provide communication between network entities **105**.

[0164] The communications manager **820** may support wireless communications at a server in accordance with examples as disclosed herein. For example, the communications manager **820** may be configured as or otherwise support a means for receiving, via a radio access network and from an aircraft, first information associated with the aircraft. The communications manager **820** may be configured as or otherwise support a means for transmitting, to the aircraft in response to receiving the first information, situational information to trigger a situational response by the aircraft while the aircraft operates in a zone associated with the server. The communications manager **820** may be configured as or otherwise support a means for communicating, between the server and an unmanned aircraft service supplier, second information associated with the aircraft, the communicating being via a network function and based on receiving the first information.

[0165] Additionally, or alternatively, the communications manager **820** may support communications at a first unmanned aircraft service supplier in accordance with examples as disclosed herein. For example, the communications manager **820** may be configured as or otherwise support a means for receiving, from a discovery and synchronization service, information associated with a second unmanned aircraft service supplier. The communications manager **820** may be configured as or otherwise support a means for receiving, from the second unmanned aircraft service supplier based on receiving the information, detect and avoid service area entity information associated with one or more detect and avoid servers managed by the second unmanned aircraft service supplier. The communications manager **820** may be configured as or otherwise support a means for transmitting, to an aircraft, an indication of the detect and avoid service area entity information that the aircraft can use while operating in an autonomous flying mode.

[0166] By including or configuring the communications manager **820** in accordance with examples as described herein, the device **805** may support techniques for improved aircraft safety and reliability by leveraging wireless communication networks to support DAA techniques.

[0167] In some examples, the communications manager **820** may be configured to perform various operations (e.g., receiving, obtaining, monitoring, outputting, transmitting) using or otherwise in cooperation with the transceiver **810**, the one or more antennas **815** (e.g., where applicable), or any combination thereof. Although the communications manager **820** is illustrated as a separate component, in some examples, one or more functions described with reference to the communications manager **820** may be supported by or performed by the processor **835**, the memory **825**, the code **830**, the transceiver **810**, or any combination thereof. For example, the code **830** may include instructions executable by the processor **835** to cause the device **805** to perform various aspects of network assisted detect and avoid systems for aerial vehicles as described herein, or the processor **835** and the memory **825** may be otherwise configured to perform or support such operations.

[0168] FIG. **9** shows a block diagram **900** of a device **905** that supports network assisted detect and avoid systems for aerial vehicles in accordance with one or more aspects of the present disclosure. The device **905** may be an example of aspects of a UE **115** as described herein. The device **905** may include a receiver **910**, a transmitter **915**, and a communications manager **920**. The device **905** may also include a processor. Each of these components may be in communication with one another (e.g., via one or more buses).

[0169] The receiver **910** may provide a means for receiving information such as packets, user data, control information, or any combination thereof associated with various information channels (e.g., control channels, data channels, information channels related to network assisted detect and avoid systems for aerial vehicles). Information may be passed on to other components of the device **905**. The receiver **910** may utilize a single antenna or a set of multiple antennas.

[0170] The transmitter **915** may provide a means for transmitting signals generated by other components of the device **905**. For example, the transmitter **915** may transmit information such as packets, user data, control information, or any combination thereof associated with various information channels (e.g., control channels, data channels, information channels related to network assisted detect and avoid systems for aerial vehicles). In some examples, the transmitter **915** may be co-located with a receiver **910** in a transceiver module. The transmitter **915** may utilize a single antenna or a set of multiple antennas.

[0171] The communications manager **920**, the receiver **910**, the transmitter **915**, or various combinations thereof or various components thereof may be examples of means for performing various aspects of network assisted detect and avoid systems for aerial vehicles as described herein. For example, the communications manager **920**, the receiver **910**, the transmitter **915**, or various combinations or components thereof may support a method for performing one or more of the functions described herein.

[0172] In some examples, the communications manager **920**, the receiver **910**, the transmitter **915**, or various combinations or components thereof may be implemented in hardware (e.g., in communications management circuitry). The hardware may include a processor, a digital signal processor (DSP), a central processing unit (CPU), an application-specific integrated circuit (ASIC), a field-programmable gate array (FPGA) or other programmable logic device, a microcontroller, discrete gate or transistor logic, discrete hardware components, or any combination thereof configured as or otherwise supporting a means for performing the functions described in the present disclosure. In some examples, a processor and memory coupled with the processor may be configured to perform one or more of the functions described herein (e.g., by executing, by the processor, instructions stored in the memory).

[0173] Additionally, or alternatively, in some examples, the communications manager **920**, the receiver **910**, the transmitter **915**, or various combinations or components thereof may be implemented in code (e.g., as communications management software or firmware) executed by a processor. If implemented in code executed by a processor, the functions of the communications manager **920**, the receiver **910**, the transmitter **915**, or various combinations or components thereof may be performed by a general-purpose processor, a DSP, a CPU, an ASIC, an FPGA, a microcontroller, or any combination of these or other programmable logic devices (e.g., configured as or otherwise supporting a means for performing the functions described in the present disclosure).

[0174] In some examples, the communications manager **920** may be configured to perform various operations (e.g., receiving, obtaining, monitoring, outputting, transmitting) using or otherwise in cooperation with the receiver **910**, the transmitter **915**, or both. For example, the communications manager **920** may receive information from the receiver **910**, send information to the transmitter **915**, or be integrated in combination with the receiver **910**, the transmitter **915**, or both to obtain information, output information, or perform various other operations as described herein.

[0175] The communications manager **920** may support wireless communications a first aircraft in accordance with examples as disclosed herein. For example, the communications manager **920** may be configured as or otherwise support a means for receiving first information associated with one or more second vehicles. The communications manager **920** may be configured as or otherwise support a means for transmitting, to a server operating in a radio access network and based on receiving the first information associated with the one or more second vehicles, second information associated with the first aircraft. The communications manager **920** may be configured as or otherwise support a means for receiving, from the server based on transmitting the second information, situational information to trigger a situational response by the first aircraft while the first aircraft operates in a zone associated with the server.

[0176] By including or configuring the communications manager **920** in accordance with examples as described herein, the device **905** (e.g., a processor controlling or otherwise coupled with the receiver **910**, the transmitter **915**, the communications manager **920**, or a combination thereof) may support techniques for improved aircraft safety and reliability by leveraging wireless communication networks to support DAA techniques.

[0177] FIG. **10** shows a block diagram **1000** of a device **1005** that supports network assisted detect and avoid systems for aerial vehicles in accordance with one or more aspects of the present disclosure. The device **1005** may be an example of aspects of a device **905** or a UE **115** as described herein. The device **1005** may include a receiver **1010**, a transmitter **1015**, and a communications manager **1020**. The device **1005** may also include a processor. Each of these components may be in communication with one another (e.g., via one or more buses).

[0178] The receiver **1010** may provide a means for receiving information such as packets, user data, control information, or any combination thereof associated with various information channels (e.g., control channels, data channels, information channels related to network assisted detect and avoid systems for aerial vehicles). Information may be passed on to other components of the device **1005**. The receiver **1010** may utilize a single antenna or a set of multiple antennas.

[0179] The transmitter **1015** may provide a means for transmitting signals generated by other components of the device **1005**. For example, the transmitter **1015** may transmit information such as packets, user data, control information, or any combination thereof associated with various information channels (e.g., control channels, data channels, information channels related to network assisted detect and avoid systems for aerial vehicles). In some examples, the transmitter **1015** may be co-located with a receiver **1010** in a transceiver module. The transmitter **1015** may utilize a single antenna or a set of multiple antennas.

[0180] The device **1005**, or various components thereof, may be an example of means for performing various aspects of network assisted detect and avoid systems for aerial vehicles as described herein. For example, the communications manager **1020** may include a second vehicle component **1025**, an aircraft information component **1030**, a situational information component **1035**, or any combination thereof. The communications manager **1020** may be an example of aspects of a communications manager **920** as described herein. In some examples, the communications manager **1020**, or various components thereof, may be configured to perform various operations (e.g., receiving, obtaining, monitoring, outputting, transmitting) using or otherwise in cooperation with the receiver **1010**, the transmitter **1015**, or both. For example, the communications manager **1020** may receive information from the receiver **1010**, send information to the transmitter **1015**, or be integrated in combination with the receiver **1010**, the transmitter **1015**, or both to obtain information, output information, or perform various other operations as described herein.

[0181] The communications manager **1020** may support wireless communications a first aircraft in accordance with examples as disclosed herein. The second vehicle component **1025** may be configured as or otherwise support a means for receiving first information associated with one or more second vehicles. The aircraft information component **1030** may be configured as or otherwise support a means for transmitting, to a server operating in a radio access network and based on receiving the first information associated with the one or more second vehicles, second information associated with the first aircraft. The situational information component **1035** may be configured as or otherwise support a means for receiving, from the server based on transmitting the second information, situational information to trigger a situational response by the first aircraft while the first aircraft operates in a zone associated with the server.

[0182] FIG. **11** shows a block diagram **1100** of a communications manager **1120** that supports network assisted detect and avoid systems for aerial vehicles in accordance with one or more aspects of the present disclosure. The communications manager **1120** may be an example of aspects of a communications manager **920**, a communications manager **1020**, or both, as described herein. The communications manager **1120**, or various components thereof, may be an example of means for performing various aspects of network assisted detect and avoid systems for aerial vehicles as described herein. For example, the communications manager **1120** may include a second vehicle component **1125**, an aircraft information component **1130**, a situational information component **1135**, an avoidance instruction component **1140**, a request or warning interface **1145**, a metadata component **1150**, an USS interface **1155**, or any combination thereof. Each of these components may communicate, directly or indirectly, with one another (e.g., via one or more buses).

[0183] The communications manager **1120** may support wireless communications a first aircraft in accordance with examples as disclosed herein. The second vehicle component **1125** may be configured as or otherwise support a means for receiving first information associated with one or more second vehicles. The aircraft information component **1130** may be configured as or otherwise support a means for transmitting, to a server operating in a radio access network and based on receiving the first information associated with the one or more second vehicles, second information associated with the first aircraft. The situational information component **1135** may be configured as or otherwise support a means for receiving, from the server based on transmitting the second information, situational information to trigger a situational response by the first aircraft while the first aircraft operates in a zone associated with the server.

[0184] In some examples, to support receiving the situational information, the situational information component **1135** may be configured as or otherwise support a means for receiving a warning indicative of a potential conflict associated with the first aircraft.

[0185] In some examples, to support receiving the situational information, the avoidance instruction component **1140** may be configured as or otherwise support a means for receiving an indication of an avoidance instruction for avoiding a potential conflict by the first aircraft.

[0186] In some examples, to support receiving the situational information, the situational information component **1135** may be configured as or otherwise support a means for receiving information associated with one or more vehicles operating in the zone or another proximal zone.

[0187] In some examples, to support receiving the situational information, the situational information component **1135** may be configured as or otherwise support a means for receiving topographical information, ground-based obstruction information, or both associated with the zone or another proximal zone.

[0188] In some examples, to support transmitting the second information, the request or warning interface **1145** may be configured as or otherwise support a means for transmitting deconfliction request or conflict warning associated with a potential conflict detected by the first aircraft, where situational information is received based on transmitting the deconfliction request or the conflict warning and includes an avoidance instruction.

[0189] In some examples, to support transmitting the second information, the metadata component **1150** may be configured as or otherwise support a means for transmitting metadata associated with the first aircraft, where the metadata includes an aircraft type, an aircraft capability, an aircraft position, an aircraft velocity, an aircraft identity, aircraft operator information, or a combination thereof.

[0190] In some examples, the aircraft information component **1130** may be configured as or otherwise support a means for transmitting, to the server, an indication of the first information associated with one or more vehicles received by the first aircraft.

[0191] In some examples, the one or more second vehicles include a piloted aircraft, an aircraft operating in the autonomous flying mode, a ground vehicle, or a combination thereof.

[0192] In some examples, the first aircraft communicates with the server via an access link, a sidelink, or both.

[0193] In some examples, the USS interface **1155** may be configured as or otherwise support a means for receiving, from an unmanned aircraft service supplier, an indication of reachability information for the server, an indication of the zone associated with the server, or both, where the first aircraft communicates with the server based on receiving the reachability information, the indication of the zone, or both.

[0194] FIG. **12** shows a diagram of a system **1200** including a device **1205** that supports network assisted detect and avoid systems for aerial vehicles in accordance with one or more aspects of the present disclosure. The device **1205** may be an example of or include the components of a device **905**, a device **1005**, or a UE **115** as described herein. The device **1205** may communicate (e.g., wirelessly) with one or more network entities **105**, one or more UEs **115**, or any combination thereof. The device **1205** may include components for bi-directional voice and data communications including components for transmitting and receiving communications, such as a communications manager **1220**, an input/output (I/O) controller **1210**, a transceiver **1215**, an antenna **1225**, a memory **1230**, code **1235**, and a processor **1240**. These components may be in electronic communication or otherwise coupled (e.g., operatively, communicatively, functionally, electronically, electrically) via one or more buses (e.g., a bus **1245**).

[0195] The I/O controller **1210** may manage input and output signals for the device **1205**. The I/O controller **1210** may also manage peripherals not integrated into the device **1205**. In some cases, the I/O controller **1210** may represent a physical connection or port to an external peripheral. In some cases, the I/O controller **1210** may utilize an operating system such as iOS®, ANDROID®, MS-DOS®, MS-WINDOWS®, OS/2®, UNIX®, LINUX®, or another known operating system. Additionally, or alternatively, the I/O controller **1210** may represent or interact with a modem, a keyboard, a mouse, a touchscreen, or a similar device. In some cases, the I/O controller **1210** may be implemented as part of a processor, such as the processor **1240**. In some cases, a user may interact with the device **1205** via the I/O controller **1210** or via hardware components controlled by the I/O controller **1210**.

[0196] In some cases, the device **1205** may include a single antenna **1225**. However, in some other cases, the device **1205** may have more than one antenna **1225**, which may be capable of concurrently transmitting or receiving multiple wireless transmissions. The transceiver **1215** may communicate bi-directionally, via the one or more antennas **1225**, wired, or wireless links as described herein. For example, the transceiver **1215** may represent a wireless transceiver and may communicate bi-directionally with another wireless transceiver. The transceiver **1215** may also include a modem to modulate the packets, to provide the modulated packets to one or more antennas **1225** for transmission, and to demodulate packets received from the one or more antennas **1225**. The transceiver **1215**, or the transceiver **1215** and one or more antennas **1225**, may be an example of a transmitter **915**, a transmitter **1015**, a receiver **910**, a receiver **1010**, or any combination thereof or component thereof, as described herein.

[0197] The memory **1230** may include random access memory (RAM) and read-only memory (ROM). The memory **1230** may store computer-readable, computer-executable code **1235** including instructions that, when executed by the processor **1240**, cause the device **1205** to perform various functions described herein. The code **1235** may be stored in a non-transitory computer-readable medium such as system memory or another type of memory. In some cases, the code **1235** may not be directly executable by the processor **1240** but may cause a computer (e.g., when compiled and executed) to perform functions described herein. In some cases, the memory **1230** may contain, among other things, a basic I/O system (BIOS) which may control basic hardware or software operation such as the interaction with peripheral components or devices.

[0198] The processor **1240** may include an intelligent hardware device (e.g., a general-purpose processor, a DSP, a CPU, a microcontroller, an ASIC, an FPGA, a programmable logic device, a discrete gate or transistor logic component, a discrete hardware component, or any combination thereof). In some cases, the processor **1240** may be configured to operate a memory array using a memory controller. In some other cases, a memory controller may be integrated into the processor **1240**. The processor **1240** may be configured to execute computer-readable instructions stored in a memory (e.g., the memory **1230**) to cause the device **1205** to perform various functions (e.g., functions or tasks supporting network assisted detect and avoid systems for aerial vehicles). For example, the device **1205** or a component of the device **1205** may include a processor **1240** and memory **1230** coupled with or to the processor **1240**, the processor **1240** and memory **1230** configured to perform various functions described herein.

[0199] The communications manager **1220** may support wireless communications a first aircraft in accordance with examples as disclosed herein. For example, the communications manager **1220** may be configured as or otherwise support a means for receiving first information associated with one or more second vehicles. The communications manager **1220** may be configured as or otherwise support a means for transmitting, to a server operating in a radio access network and based on receiving the first information associated with the one or more second vehicles, second information associated with the first aircraft. The communications manager **1220** may be configured as or otherwise support a means for receiving, from the server based on transmitting the second information, situational information to trigger a situational response by the first aircraft while the first aircraft operates in a zone associated with the server.

[0200] By including or configuring the communications manager **1220** in accordance with examples as described herein, the device **1205** may support techniques for improved aircraft safety and reliability by leveraging wireless communication networks to support DAA techniques.

[0201] In some examples, the communications manager **1220** may be configured to perform various operations (e.g., receiving, monitoring, transmitting) using or otherwise in cooperation with the transceiver **1215**, the one or more antennas **1225**, or any combination thereof. Although the communications manager **1220** is illustrated as a separate component, in some examples, one or more functions described with reference to the communications manager **1220** may be supported by or performed by the processor **1240**, the memory **1230**, the code **1235**, or any combination thereof. For example, the code **1235** may include instructions executable by the processor **1240** to cause the device **1205** to perform various aspects of network assisted detect and avoid systems for aerial vehicles as described herein, or the processor **1240** and the memory **1230** may be otherwise configured to perform or support such operations.

[0202] FIG. **13** shows a flowchart illustrating a method **1300** that supports network assisted detect and avoid systems for aerial vehicles in accordance with one or more aspects of the present disclosure. The operations of the method **1300** may be implemented by a network entity or its components as described herein. For example, the operations of the method **1300** may be performed by a network entity as described with reference to FIGS. **1** through **8**. In some examples, a network entity may execute a set of instructions to control the functional elements of the network entity to perform the described functions. Additionally, or alternatively, the network entity may perform aspects of the described functions using special-purpose hardware.

[0203] At **1305**, the method may include receiving, via a radio access network and from an aircraft, first information associated with the aircraft. The operations of **1305** may be performed in accordance with examples as disclosed herein. In some examples, aspects of the operations of **1305** may be performed by an aircraft information component **725** as described with reference to FIG. **7**.

[0204] At **1310**, the method may include transmitting, to the aircraft in response to receiving the first information, situational information to trigger a situational response by the aircraft while the aircraft operates in a zone associated with the server. The operations of **1310** may be performed in accordance with examples as disclosed herein. In some examples, aspects of the operations of **1310** may be performed by a situational information component **730** as described with reference to FIG.

7.

[0205] At **1315**, the method may include communicating, between the server and an unmanned aircraft service supplier, second information associated with the aircraft, the communicating being via a network function and based at least in part on receiving the first information. The operations of **1315** may be performed in accordance with examples as disclosed herein. In some examples, aspects of the operations of **1315** may be performed by an USS interface **735** as described with reference to FIG. **7**.

[0206] FIG. **14** shows a flowchart illustrating a method **1400** that supports network assisted detect and avoid systems for aerial vehicles in accordance with one or more aspects of the present disclosure. The operations of the method **1400** may be implemented by a UE or its components as described herein. For example, the operations of the method **1400** may be performed by a UE **115** as described with reference to FIGS. **1** through **4** and **9** through **12**. In some examples, a UE may execute a set of instructions to control the functional elements of the UE to perform the described functions. Additionally, or alternatively, the UE may perform aspects of the described functions using special-purpose hardware.

[0207] At **1405**, the method may include receiving first information associated with one or more second vehicles. The operations of **1405** may be performed in accordance with examples as disclosed herein. In some examples, aspects of the operations of **1405** may be performed by a second vehicle component **1125** as described with reference to FIG. **11**.

[0208] At **1410**, the method may include transmitting, to a server operating in a radio access network and based at least in part on receiving the first information associated with the one or more second vehicles, second information associated with the first aircraft. The operations of **1410** may be performed in accordance with examples as disclosed herein. In some examples, aspects of the operations of **1410** may be performed by an aircraft information component **1130** as described with reference to FIG. **11**.

[0209] At **1415**, the method may include receiving, from the server based at least in part on transmitting the second information, situational information to trigger a situational response by the first aircraft while the first aircraft operates in a zone associated with the server. The operations of **1415** may be performed in accordance with examples as disclosed herein. In some examples, aspects of the operations of **1415** may be performed by a situational information component **1135** as described with reference to FIG. **11**.

[0210] FIG. **15** shows a flowchart illustrating a method **1500** that supports network assisted detect and avoid systems for aerial vehicles in accordance with one or more aspects of the present disclosure. The operations of the method **1500** may be implemented by a network entity or its components as described herein. For example, the operations of the method **1500** may be performed by a network entity as described with reference to FIGS. **1** through **8**. In some examples, a network entity may execute a set of instructions to control the functional elements of the network entity to perform the described functions. Additionally, or alternatively, the network entity may perform aspects of the described functions using special-purpose hardware.

[0211] At **1505**, the method may include receiving, from a discovery and synchronization service, information associated with a second unmanned aircraft service supplier. The operations of **1505** may be performed in accordance with examples as disclosed herein. In some examples, aspects of the operations of **1505** may be performed by a DSS interface **740** as described with reference to FIG. **7**.

[0212] At **1510**, the method may include receiving, from the second unmanned aircraft service supplier based at least in part on receiving the information, detect and avoid service area entity information associated with one or more detect and avoid servers managed by the second unmanned aircraft service supplier. The operations of **1510** may be performed in accordance with examples as disclosed herein. In some examples, aspects of the operations of **1510** may be performed by an USS interface **735** as described with reference to FIG. **7**.

[0213] At **1515**, the method may include transmitting, to an aircraft, an indication of the detect and avoid service area entity information that the aircraft can use while operating in an autonomous flying mode. The operations of **1515** may be performed in accordance with examples as disclosed herein. In some examples, aspects of the operations of **1515** may be performed by an aircraft interface **745** as described with reference to FIG. **7**.

[0214] The following provides an overview of aspects of the present disclosure:

[0215] Aspect 1: A method for wireless communications at a server, comprising: receiving, via a radio access network and from an aircraft, first information associated with the aircraft; transmitting, to the aircraft in response to receiving the first information, situational information to trigger a situational response by the aircraft while the aircraft operates in a zone associated with the server; and communicating, between the server and an unmanned aircraft service supplier, second information associated with the aircraft, the communicating being via a network function and based at least in part on receiving the first information.

[0216] Aspect 2: The method of aspect 1, wherein transmitting the situational information comprises: transmitting a warning indicative of a potential conflict associated with the aircraft.

[0217] Aspect 3: The method of any of aspects 1 through 2, wherein transmitting the situational information comprises: transmitting an indication of an avoidance instruction for avoiding a potential conflict by the aircraft.

[0218] Aspect 4: The method of any of aspects 1 through 3, wherein transmitting the situational information comprises: transmitting neighboring vehicle information associated with one or more vehicles operating in the zone or another proximal zone.

[0219] Aspect 5: The method of any of aspects 1 through 4, wherein transmitting the situational information comprises: transmit topographical information, ground-based obstruction information, or both associated with the zone or another proximal zone.

[0220] Aspect 6: The method of any of aspects 1 through 5, wherein receiving the first information comprises: receiving a deconfliction request or conflict warning associated with a potential conflict detected by the aircraft, wherein situational information is transmitted based at least in part on receiving the deconfliction request or the conflict warning and includes an avoidance instruction.

[0221] Aspect 7: The method of any of aspects 1 through 6, wherein receiving the first information comprises: receiving metadata associated with the aircraft, wherein the metadata comprises an aircraft type, an aircraft capability, an aircraft position, an aircraft velocity, an aircraft identity, aircraft operator information, or a combination thereof.

[0222] Aspect 8: The method of any of aspects 1 through 7, further comprising: receiving, from the aircraft, third information associated with one or more vehicles detected by the aircraft.

[0223] Aspect 9: The method of aspect 8, wherein the one or more vehicles comprise a piloted aircraft, an aircraft operating in the autonomous flying mode, a ground vehicle, or a combination thereof.

[0224] Aspect 10: The method of any of aspects 1 through 9, wherein communicating the second information comprises: transmitting, to the unmanned aircraft service supplier, an indication of conflict information associated with the aircraft, an indication of vehicle congestion information of the zone, or a combination thereof.

[0225] Aspect 11: The method of any of aspects 1 through 10, wherein communicating the second information comprises: transmitting, to the unmanned aircraft service supplier, a request for the second information associated with the aircraft; and receiving, from the unmanned aircraft service supplier, in response to transmitting the request, the second information.

[0226] Aspect 12: The method of any of aspects 1 through 11, further comprising: determining, based at least in part on receiving the first information, that a potential conflict exists for the aircraft, a second aircraft, or both; and transmitting, to the aircraft, the second aircraft, or both, an indication of the potential conflict.

[0227] Aspect 13: The method of any of aspects 1 through 12, further comprising: receiving, from one or more sensors associated with the server, zone information associated with the zone, wherein the situational information is based at least in part on the zone information.

[0228] Aspect 14: The method of aspect 13, wherein the one or more sensors comprise a detect and avoid broadcast receiver, a broadcast remote identifier receiver, an automatic dependent surveillance-broadcast receiver, a weather sensor, a radar, a new-radio sensor, a lidar, an aircraft transponder, or a combination thereof.

[0229] Aspect 15: The method of any of aspects 1 through 14, wherein the server communicates with the aircraft via an access link, a sidelink, or both.

[0230] Aspect 16: A method for wireless communications a first aircraft, comprising: receiving first information associated with one or more second vehicles; transmitting, to a server operating in a radio access network and based at least in part on receiving the first information associated with the one or more second vehicles, second information associated with the first aircraft; and receiving, from the server based at least in part on transmitting the second information, situational information to trigger a situational response by the first aircraft while the first aircraft operates in a zone associated with the server.

[0231] Aspect 17: The method of aspect 16, wherein receiving the situational information comprises: receiving a warning indicative of a potential conflict associated with the first aircraft.

[0232] Aspect 18: The method of any of aspects 16 through 17, wherein receiving the situational information comprises: receiving an indication of an avoidance instruction for avoiding a potential conflict by the first aircraft.

[0233] Aspect 19: The method of any of aspects 16 through 18, wherein receiving the situational information comprises: receiving information associated with one or more vehicles operating in the zone or another proximal zone.

[0234] Aspect 20: The method of any of aspects 16 through 19, wherein receiving the situational information comprises: receive topographical information, ground-based obstruction information, or both associated with the zone or another proximal zone.

[0235] Aspect 21: The method of any of aspects 16 through 20, wherein transmitting the second information comprises: transmitting deconfliction request or conflict warning associated with a potential conflict detected by the first aircraft, wherein situational information is received based at least in part on transmitting the deconfliction request or the conflict warning and includes an avoidance instruction.

[0236] Aspect 22: The method of any of aspects 16 through 21, wherein transmitting the second information comprises: transmitting metadata associated with the first aircraft, wherein the metadata comprises an aircraft type, an aircraft capability, an aircraft position, an aircraft velocity, an aircraft identity, aircraft operator information, or a combination thereof.

[0237] Aspect 23: The method of any of aspects 16 through 22, further comprising: transmitting, to the server, an indication of the first information associated with one or more vehicles received by the first aircraft.

[0238] Aspect 24: The method of any of aspects 16 through 23, wherein the one or more second vehicles comprise a piloted aircraft, an aircraft operating in an autonomous flying mode, a ground vehicle, or a combination thereof.

[0239] Aspect 25: The method of any of aspects 16 through 24, wherein the first aircraft communicates with the server via an access link, a sidelink, or both.

[0240] Aspect 26: The method of any of aspects 16 through 25, further comprising: receiving, from an unmanned aircraft service supplier, an indication of reachability information for the server, an indication of the zone associated with the server, or both, wherein the first aircraft communicates with the server based at least in part on receiving the reachability information, the indication of the zone, or both.

[0241] Aspect 27: A method for communications at a first unmanned aircraft service supplier, comprising: receiving, from a discovery and synchronization service, information associated with a second unmanned aircraft service supplier; receiving, from the second unmanned aircraft service supplier based at least in part on receiving the information, detect and avoid service area entity information associated with one or more detect and avoid servers managed by the second unmanned aircraft service supplier; and transmitting, to an aircraft, an indication of the detect and avoid service area entity information that the aircraft is use.

[0242] Aspect 28: The method of aspect 27, wherein receiving the information associated with the second unmanned aircraft service supplier comprises: receiving an indication of reachability information for the one or more detected and avoid servers, a location of the one or more detect avoid servers, a zone identifier associated with the one or more detect and avoid servers, or a combination thereof.

[0243] Aspect 29: The method of any of aspects 27 through 28, further comprising: transmitting, to the second unmanned aircraft service supplier in response to receiving the information, a request for the detect and avoid service area entity information, wherein the detect and avoid service area entity information is received based at least in part on transmitting the request.

[0244] Aspect 30: An apparatus for wireless communications at a server, comprising a processor; memory coupled with the processor; and instructions stored in the memory and executable by the processor to cause the apparatus to perform a method of any of aspects 1 through 15.

[0245] Aspect 31: An apparatus for wireless communications at a server, comprising at least one means for performing a method of any of aspects 1 through 15.

[0246] Aspect 32: A non-transitory computer-readable medium storing code for wireless communications at a server, the code comprising instructions executable by a processor to perform a method of any of aspects 1 through 15.

[0247] Aspect 33: An apparatus for wireless communications a first aircraft, comprising a processor; memory coupled with the processor; and instructions stored in the memory and executable by the processor to cause the apparatus to perform a method of any of aspects 16 through 26.

[0248] Aspect 34: An apparatus for wireless communications a first aircraft, comprising at least one means for performing a method of any of aspects 16 through 26.

[0249] Aspect 35: A non-transitory computer-readable medium storing code for wireless communications a first aircraft, the code comprising instructions executable by a processor to perform a method of any of aspects 16 through 26.

[0250] Aspect 36: An apparatus for communications at a first unmanned aircraft service supplier, comprising a processor; memory coupled with the processor; and instructions stored in the memory and executable by the processor to cause the apparatus to perform a method of any of aspects 27 through 29.

[0251] Aspect 37: An apparatus for communications at a first unmanned aircraft service supplier, comprising at least one means for performing a method of any of aspects 27 through 29.

[0252] Aspect 38: A non-transitory computer-readable medium storing code for communications at a first unmanned aircraft service supplier, the code comprising instructions executable by a processor to perform a method of any of aspects 27 through 29.

[0253] It should be noted that the methods described herein describe possible implementations, and that the operations and the steps may be rearranged or otherwise modified and that other implementations are possible. Further, aspects from two or more of the methods may be combined.

[0254] Although aspects of an LTE, LTE-A, LTE-A Pro, or NR system may be described for purposes of example, and LTE, LTE-A, LTE-A Pro, or NR terminology may be used in much of the description, the techniques described herein are applicable beyond LTE, LTE-A, LTE-A Pro, or NR networks. For example, the described techniques may be applicable to various other wireless communications systems such as Ultra Mobile Broadband (UMB), Institute of Electrical and Electronics Engineers (IEEE) 802.11 (Wi-Fi), IEEE 802.16 (WiMAX), IEEE 802.20, Flash-OFDM, as well as other systems and radio technologies not explicitly mentioned herein.

[0255] Information and signals described herein may be represented using any of a variety of different technologies and techniques. For example, data, instructions, commands, information, signals, bits, symbols, and chips that may be referenced throughout the description may be represented by voltages, currents, electromagnetic waves, magnetic fields or particles, optical fields or particles, or any combination thereof.

[0256] The various illustrative blocks and components described in connection with the disclosure herein may be implemented or performed using a general-purpose processor, a DSP, an ASIC, a CPU, an FPGA or other programmable logic device, discrete gate or transistor logic, discrete hardware components, or any combination thereof designed to perform the functions described herein. A general-purpose processor may be a microprocessor but, in the alternative, the processor may be any processor, controller, microcontroller, or state machine. A processor may also be implemented as a combination of computing devices (e.g., a combination of a DSP and a microprocessor, multiple microprocessors, one or more microprocessors in conjunction with a DSP core, or any other such configuration).

[0257] The functions described herein may be implemented using hardware, software executed by a processor, firmware, or any combination thereof. If implemented using software executed by a processor, the functions may be stored as or transmitted using one or more instructions or code of a computer-readable medium. Other examples and implementations are within the scope of the disclosure and appended claims. For example, due to the nature of software, functions described herein may be implemented using software executed by a processor, hardware, firmware, hardwiring, or combinations of any of these. Features implementing functions may also be physically located at various positions, including being distributed such that portions of functions are implemented at different physical locations.

[0258] Computer-readable media includes both non-transitory computer storage media and communication media including any medium that facilitates transfer of a computer program from one location to another. A non-transitory storage medium may be any available medium that may be accessed by a general-purpose or special-purpose computer. By way of example, and not limitation, non-transitory computer-readable media may include RAM, ROM, electrically erasable programmable ROM (EEPROM), flash memory, compact disk (CD) ROM or other optical disk storage, magnetic disk storage or other magnetic storage devices, or any other non-transitory medium that may be used to carry or store desired program code means in the form of instructions or data structures and that may be accessed by a general-purpose or special-purpose computer, or a general-purpose or special-purpose processor. Also, any connection is properly termed a computer-readable medium. For example, if the software is transmitted from a website, server, or other remote source using a coaxial cable, fiber optic cable, twisted pair, digital subscriber line (DSL), or wireless technologies such as infrared, radio, and microwave, then the coaxial cable, fiber optic cable, twisted pair, DSL, or wireless technologies such as infrared, radio, and microwave are included in the definition of computer-readable medium. Disk and disc, as used herein, include CD, laser disc, optical disc, digital versatile disc (DVD), floppy disk and Blu-ray disc. Disks may reproduce data magnetically, and discs may reproduce data optically using lasers. Combinations of the above are also included within the scope of computer-readable media.

[0259] As used herein, including in the claims, “or” as used in a list of items (e.g., a list of items prefaced by a phrase such as “at least one of” or “one or more of”) indicates an inclusive list such that, for example, a list of at least one of A, B, or C means A or B or C or AB or AC or BC or ABC (i.e., A and B and C). Also, as used herein, the phrase “based on” shall not be construed as a reference to a closed set of conditions. For example, an example step that is described as “based on condition A” may be based on both a condition A and a condition B without departing from the scope of the present disclosure. In other words, as used herein, the phrase “based on” shall be construed in the same manner as the phrase “based at least in part on.”

[0260] The term “determine” or “determining” encompasses a variety of actions and, therefore, “determining” can include calculating, computing, processing, deriving, investigating, looking up (such as via looking up in a table, a database or another data structure), ascertaining and the like. Also, “determining” can include receiving (e.g., receiving information), accessing (e.g., accessing data stored in memory) and the like. Also, “determining” can include resolving, obtaining, selecting, choosing, establishing, and other such similar actions.

[0261] In the appended figures, similar components or features may have the same reference label. Further, various components of the same type may be distinguished by following the reference label by a dash and a second label that distinguishes among the similar components. If just the first reference label is used in the specification, the description is applicable to any one of the similar components having the same first reference label irrespective of the second reference label, or other subsequent reference label.

[0262] The description set forth herein, in connection with the appended drawings, describes example configurations and does not represent all the examples that may be implemented or that are within the scope of the claims. The term “example” used herein means “serving as an example, instance, or illustration,” and not “preferred” or “advantageous over other examples.” The detailed description includes specific details for the purpose of providing an understanding of the described techniques. These techniques, however, may be practiced without these specific details. In some instances, known structures and devices are shown in block diagram form in order to avoid obscuring the concepts of the described examples.

[0263] The description herein is provided to enable a person having ordinary skill in the art to make or use the disclosure. Various modifications to the disclosure will be apparent to a person having ordinary skill in the art, and the generic principles defined herein may be applied to other variations without departing from the scope of the disclosure. Thus, the disclosure is not limited to the examples and designs described herein but is to be accorded the broadest scope consistent with the principles and novel features disclosed herein.

Claims

1. An apparatus for wireless communications at a server, comprising: a processor; memory coupled with the processor; and instructions stored in the memory and executable by the processor to cause the apparatus to: receive, via a radio access network and from an aircraft, first information associated with the aircraft; transmit, to the aircraft in response to receiving the first information, situational information to trigger a situational response by the aircraft while the aircraft operates in a zone associated with the server; and communicate, between the server and an unmanned aircraft service supplier, second information associated with the aircraft, the communicating being via a network function and based at least in part on receiving the first information.

2. The apparatus of claim 1, wherein the instructions to transmit the situational information are executable by the processor to cause the apparatus to: transmit a warning indicative of a potential conflict associated with the aircraft.

3. The apparatus of claim 1, wherein the instructions to transmit the situational information are executable by the processor to cause the apparatus to: transmit an indication of an avoidance instruction for avoiding a potential conflict by the aircraft.

4. The apparatus of claim 1, wherein the instructions to transmit the situational information are executable by the processor to cause the apparatus to: transmit neighboring vehicle information associated with one or more vehicles operating in the zone or another proximal zone.

5. The apparatus of claim 1, wherein the instructions to transmit the situational information are executable by the processor to cause the apparatus to: transmit topographical information, ground-based obstruction information, or both associated with the zone or another proximal zone.

6. The apparatus of claim 1, wherein the instructions to receive the first information are executable by the processor to cause the apparatus to: receive a deconfliction request or conflict warning

associated with a potential conflict detected by the aircraft, wherein situational information is transmitted based at least in part on receiving the deconfliction request or the conflict warning and includes an avoidance instruction.
7. The apparatus of claim 1, wherein the instructions to receive the first information are executable by the processor to cause the apparatus to: receive metadata associated with the aircraft, wherein the metadata comprises an aircraft type, an aircraft capability, an aircraft position, an aircraft velocity, an aircraft identity, aircraft operator information, or a combination thereof.
8. The apparatus of claim 1, wherein the instructions are further executable by the processor to cause the apparatus to: receive, from the aircraft, third information associated with one or more vehicles detected by the aircraft.
9. The apparatus of claim 8, wherein the one or more vehicles comprise a piloted aircraft, an aircraft operating in an autonomous flying mode, a ground vehicle, or a combination thereof.
10. The apparatus of claim 1, wherein the instructions to communicate the second information are executable by the processor to cause the apparatus to: transmit, to the unmanned aircraft service supplier, an indication of conflict information associated with the aircraft, an indication of vehicle congestion information of the zone, or a combination thereof.
11. The apparatus of claim 1, wherein the instructions to communicate the second information are executable by the processor to cause the apparatus to: transmit, to the unmanned aircraft service supplier, a request for the second information associated with the aircraft; and receive, from the unmanned aircraft service supplier, in response to transmitting the request, the second information.
12. The apparatus of claim 1, wherein the instructions are further executable by the processor to cause the apparatus to: determine, based at least in part on receiving the first information, that a potential conflict exists for the aircraft, a second aircraft, or both; and transmit, to the aircraft, the second aircraft, or both, an indication of the potential conflict.
13. The apparatus of claim 1, wherein the instructions are further executable by the processor to cause the apparatus to: receive, from one or more sensors associated with the server, zone information associated with the zone, wherein the situational information is based at least in part on the zone information.
14. The apparatus of claim 13, wherein the one or more sensors comprise a detect and avoid broadcast receiver, a broadcast remote identifier receiver, an automatic dependent surveillance-broadcast receiver, a weather sensor, a radar, a new-radio sensor, a lidar, an aircraft transponder, or a combination thereof.
15. The apparatus of claim 1, wherein the server communicates with the aircraft via an access link, a sidelink, or both.
16. An apparatus for wireless communications a first aircraft, comprising: a processor; memory coupled with the processor; and instructions stored in the memory and executable by the processor to cause the apparatus to: receive first information associated with one or more second vehicles; transmit, to a server operating in a radio access network and based at least in part on receiving the first information associated with the one or more second vehicles, second information associated with the first aircraft; and receive, from the server based at least in part on transmitting the second information, situational information to trigger a situational response by the first aircraft while the first aircraft operates in a zone associated with the server.
17. The apparatus of claim 16, wherein the instructions to receive the situational information are executable by the processor to cause the apparatus to: receive a warning indicative of a potential conflict associated with the first aircraft.
18. The apparatus of claim 16, wherein the instructions to receive the situational information are executable by the processor to cause the apparatus to: receive an indication of an avoidance instruction for avoiding a potential conflict by the first aircraft.
19. The apparatus of claim 16, wherein the instructions to receive the situational information are executable by the processor to cause the apparatus to: receive information associated with one or

more vehicles operating in the zone or another proximal zone.
20. The apparatus of claim 16, wherein the instructions to receive the situational information are executable by the processor to cause the apparatus to: receive topographical information, ground-based obstruction information, or both associated with the zone or another proximal zone.
21. The apparatus of claim 16, wherein the instructions to transmit the second information are executable by the processor to cause the apparatus to: transmit deconfliction request or conflict warning associated with a potential conflict detected by the first aircraft, wherein situational information is received based at least in part on transmitting the deconfliction request or the conflict warning and includes an avoidance instruction.
22. The apparatus of claim 16, wherein the instructions to transmit the second information are executable by the processor to cause the apparatus to: transmit metadata associated with the first aircraft, wherein the metadata comprises an aircraft type, an aircraft capability, an aircraft position, an aircraft velocity, an aircraft identity, aircraft operator information, or a combination thereof.
23. The apparatus of claim 16, wherein the instructions are further executable by the processor to cause the apparatus to: transmit, to the server, an indication of the first information associated with one or more vehicles received by the first aircraft.
24. The apparatus of claim 16, wherein the one or more second vehicles comprise a piloted aircraft, an aircraft operating in an autonomous flying mode, a ground vehicle, or a combination thereof.
25. The apparatus of claim 16, wherein: the first aircraft communicates with the server via an access link, a sidelink, or both.
26. The apparatus of claim 16, wherein the instructions are further executable by the processor to cause the apparatus to: receive, from an unmanned aircraft service supplier, an indication of reachability information for the server, an indication of the zone associated with the server, or both, wherein the first aircraft communicates with the server based at least in part on receiving the reachability information, the indication of the zone, or both.
27. A method for wireless communications at a server, comprising: receiving, via a radio access network and from an aircraft, first information associated with the aircraft; transmitting, to the aircraft in response to receiving the first information, situational information to trigger a situational response by the aircraft while the aircraft operates in a zone associated with the server; and communicating, between the server and an unmanned aircraft service supplier, second information associated with the aircraft, the communicating being via a network function and based at least in part on receiving the first information.
28. The method of claim 27, wherein transmitting the situational information comprises: transmitting an indication of an avoidance instruction for avoiding a potential conflict by the aircraft.
29. The method of claim 27, wherein receiving the first information comprises: receiving a deconfliction request or conflict warning associated with a potential conflict detected by the aircraft, wherein situational information is transmitted based at least in part on receiving the deconfliction request or the conflict warning and includes an avoidance instruction.
30. A method for wireless communications a first aircraft, comprising: receiving first information associated with one or more second vehicles; transmitting, to a server operating in a radio access network and based at least in part on receiving the first information associated with the one or more second vehicles, second information associated with the first aircraft; and receiving, from the server based at least in part on transmitting the second information, situational information to trigger a situational response by the first aircraft while the first aircraft operates in a zone associated with the server.